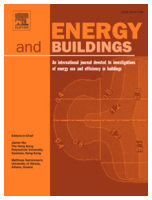
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Commercial building HVAC demand flexibility with model predictive control: Field demonstration and literature insights

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ABSTRACT

Model Predictive Control (MPC) for building Heating Ventilation and Air Conditioning (HVAC) systems is beginning to gain traction in the market, with a few controls companies incorporating it into their product offerings. However, it remains difficult to assess whether the energy cost savings are enough to justify the cost of MPC implementation for a particular building, given the limited number of reported demonstrations. For small commercial and residential buildings with relatively uniform systems, standardized approaches can help lower implementation costs. In contrast, for large buildings or district systems, the potential magnitude of cost savings could justify more customized solutions. Estimating the cost-effectiveness of MPC becomes more challenging for medium and large commercial buildings, where a one-size-fits-all solution may not be suitable, and the potential energy cost savings may be insufficient to justify a customized solution. To make MPC technology more appealing, incorporating additional value streams beyond energy efficiency alone can significantly increase its attractiveness. One such revenue stream is demand flexibility, in response to dynamic electricity prices, where MPC can leverage the thermal mass of the building to shift the load and support the grid. Building on an extensive literature review of MPC field studies focused on cost savings and demand flexibility, this paper presents the results of implementing MPC control in a large office building HVAC system in Berkeley, CA. Four different dynamic electricity price profiles were integrated into the MPC objective function to shift building demand while maintaining comfort, and field testing was performed with each price profile across four seasons. The results show potential for 40–65 % demand decrease percentage and up to 61 % annual cost savings compared to the existing rule-based control strategy, under the tested dynamic price scenarios. This paper also presents a sensitivity analysis on the cost savings with respect to the price profile variability, discusses the implementation effort for the price-responsive MPC, and compares the cost savings found in this study to those found in literature on the basis of dynamic price variability, or so-called Electricity Price Relative Standard Deviation.

1. Introduction

1.1. Background

Buildings will play an important role in providing electrical demand flexibility to help lower operating costs and improve reliability of the electric grid [1]. However, buildings will require more advanced load control than conventionally used to provide such flexibility. They currently account for more than 70 % of the electricity use in the U.S., making buildings a key consideration for demand flexibility. Buildings can provide demand flexibility in a variety of ways [2], including shifting HVAC load in response to Demand Response (DR) event signals or dynamic electricity prices that discourage electricity use during peak hours, therefore not only providing flexibility for the grid but also saving energy costs for the building owner. Passive thermal energy storage in building envelope and internal mass (e.g. furniture, partitions) provides an opportunity for HVAC systems to operate flexibly without compromising thermal comfort, and without the need for purchasing additional active energy storage, such as a battery or water or ice tank. Furthermore, commercial buildings account for 47 % of building energy use in the U.S. [3]. Large commercial buildings (those greater than 50,000 sf), account for 55 % of commercial building energy usage, despite only accounting for 6 % of the number of commercial buildings [4]. Finally, in large commercial buildings, heating, cooling, and ventilation account for approximately half of their energy use on average [5]. These figures indicate large commercial building HVAC systems to be good targets

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Nomenclature

$\boldsymbol{\omega}$	MPC model disturbance vector
$\boldsymbol{\theta}_e^*$	MPC model parameters used for control optimization
$\boldsymbol{\theta}^l$	MPC model parameter lower limits
$\boldsymbol{\theta}^u$	MPC model parameter upper limits
$\boldsymbol{\theta}_e$	Parameters to be estimated
$\boldsymbol{\theta}_f$	MPC model parameters that are fixed
$\tilde{\boldsymbol{\omega}}$	MPC model forecasted disturbance vector
ΔX	uncertainty in X
$\dot{m}_{oa}^l$	lower limit on outside air flow rate
$\dot{m}_{sa}^l$	lower limit on supply air flow rate
$\dot{m}_{0,sf}$	supply fan nominal air flow rate
$\dot{m}_{sf}$	supply fan air flow rate
$\hat{y}$	MPC model output data
$\dot{\mathbf{x}}$	MPC model state vector time derivative
$\mathbf{u}$	MPC model control input vector
$\mathbf{x}$	MPC model state vector
$\mathbf{x}_0^*$	MPC model state vector initial condition used by control optimization
$\mathbf{x}_0$	MPC model state vector initial condition
c_p	specific heat capacity of air
COP_{wshp}	estimated WSHP COP
d	day indicator
d	number of days
f	MPC model
f_{cp}	change point regression model
J	objective variable
k	control regularization weights
n	sample size
P	MPC model sum of supply and return fan power, DX power, UFT reheat power
$P_{0,rf}$	return fan nominal power
$P_{0,sf}$	supply fan nominal power
P_{dx}	DX power
P_{hvac1}	HVAC panel 1 power measurement
P_{hvac2}	HVAC panel 2 power measurement
P_{rf}	return fan power
P_{sf}	supply fan power
Q_{wshw}	reheat water heat flow rate measured at WSHP
R^2	coefficient of determination
R^n	Set of real numbers with dimension n
s	slack variables
T_z^c	zone cooling set point
T_z^h	zone heating set point
T_z^l	lower constraint on zone air temperature
T_{sa}^l	lower constraint on supply air temperature
T_z^u	upper constraint on zone air temperature
T_{sa}^u	upper constraint on supply air temperature
t_0	start time
t_f	final time
T_z	zone air temperature
T_{ma}	mixed air temperature
$T_{mdb,b}$	mean daily outdoor air dry-bulb temperature
T_{sa}	supply air temperature
u_{dx}	DX control signal
u_{hw}	UFT hot water valve control signal
u_{oa}	outside air damper control signal
u_{sf}	supply fan control signal
w	slack variable weights
y	measurement data
Z	total number of zones
z	Zone indicator

Abbrivation

AHU	air handling unit
ALC	automated logic corporation
API	application programming interface
ARX	autoregressive model with exogenous input
BMS	building management system
COP	coefficient of performance
DX	direct expansion
EMIS	energy management information system
EPRSD	electricity price relative standard deviation
EUI	energy use intensity
FCU	fan coil unit
GHI	global horizontal irradiation
HP	heat pump
HVAC	heating, ventilation, and air conditioning
LBNL	Lawrence Berkeley National Laboratory
MPC	model predictive control
MSE	mean square error
NERSC	national energy research scientific computing center
NOAA	national oceanic and atmospheric administration
PI	proportional-integral
PV	photovoltaic system
RC	resistance-capacitance
RMSE	root mean square error
RSD	relative standard deviation
RTU	rooftop unit
TABS	thermally activated building structures
TSO	transmission system operator
UFAD	underfloor air distribution
UFT	underfloor terminal unit
VAV	variable air volume
VFD	variable frequency drive

for energy efficiency and load flexibility investments, with the ability to make a significant impact per building.

Model Predictive Control (MPC) can be an effective solution to control the HVAC system to take advantage of the building envelope and internal mass to shift electricity use, by predicting future building behavior based on a model of the building and forecasts for weather and variable electricity price, or other demand flexibility incentives [6], while maintaining comfort. In fact there has been a steady rise in the number of simulation and experimental studies on MPC in the last two decades, and companies are emerging aiming to offer MPC. However, despite its potential value and the growing interest by industry, several challenges must still be addressed to promote widespread adoption, especially in large commercial buildings. This paper focuses on addressing challenges related to the cost-benefit analysis of MPC, especially in the context of providing grid flexibility services. It utilizes results from a recent field demonstration of MPC providing price-based demand flexibility in a large commercial building, along with findings from recent literature.

1.2. Literature and challenges

We reviewed 19 experimental studies implementing MPC in real commercial buildings or test chambers, all of which provided some form of demand flexibility scenario. The review was carried out using search channels like Google Scholar and Scopus, as well as general search through institutional technical reports and conference proceedings starting from 1990 to 2024. Targeted keywords were used “Model Predictive Control,” “commercial buildings,” “cost savings,” and “building

automation" to capture a wide range of studies, and a snowballing approach was used after. The results are presented in detail in Table 1. Previous review work [7] focusing on MPC experimental studies for commercial buildings found that there is a wide variation in energy and cost savings. The manuscript attributes this variation to the length and scope of the tests. Longer tests tend to result in smaller savings and whole-building tests yield smaller savings than partial-building tests. This may be due to unaccounted coupling between zones, where savings achieved in one part of the building are actually transferring load to another part. For studies that focused on the whole-building implementations, the manuscript reports average MPC energy savings of 12.9 % and average cost savings of 24.7 %. Using the papers identified from the literature, we conducted a meta-analysis on the cost savings for whole-building studies present in Khabbazi et al. [7]. The analysis revealed a ±12 % standard deviation in reported cost savings, indicating significant variability compared to the 24.7 % average. From our literature review, two additional significant factors seem to contribute to the variability in cost savings: 1) the system's flexibility potential, which refers to the control opportunities that MPC can leverage to introduce flexibility, and which the baseline control solution cannot effectively manage; and 2) the flexibility incentive, or the economic incentives provided to induce flexibility.

Regarding the first factor, the system's flexibility potential, consider three studies from our review. In [8], a battery was controlled to maximize photovoltaic PV self-consumption and shift load away from high electricity prices. The study achieved 2 % cost savings with a battery sized to shift 15 % of the daily load and 11 % with a battery sized to shift 30 % of the daily load. In [9], both a battery and a thermostat set points were controlled to leverage the building thermal mass and shift load away from high electricity prices. The authors report 12 % cost savings with the battery sized for approximately 20 % of the daily load and the HVAC load accounted for 15–25 % of the total electrical load. In [10], a battery and heat pump were managed in a new university building to shift load away from high electricity prices and provide ancillary demand response services. This led to 28 % cost savings, even if the battery was sized only for 5 % of the daily load, while HVAC contributed the majority of the electrical load.

As for flexibility incentives, the primary economic drivers in the reviewed studies were variable energy prices ($/kWh) and/or demand charges ($/kW). Variable prices come in several forms, such as Time of Use (TOU) prices, where prices change in steps throughout the day or the season , and other highly dynamic prices that can vary hourly or sub-hourly, for example, based on the electricity whole-sale market spot price. In our review, 11 studies addressed TOU prices, while two studies responded to highly dynamic prices. Demand charges, often applied monthly on a customer's peak demand, were another factor. These can also vary based on different periods in the day. Four studies aimed to minimize peak demand or responded to demand charges. Additional incentives included maximizing PV self-consumption [11], responding to different prices for electricity import and export [8,12], and providing grid services such as regulation, reserve or load-tracking [13–15].

A few papers reported the implementation costs for MPC, which is critical factor for conducting a comprehensive cost-benefit analysis. In 2022, Blum et al. [16] highlighted that most studies in the literature until then primarily aimed to demonstrate MPC technology across different scenarios, with only one study [17] reporting implementation costs. Blum et al. [16] addressed this gap by presenting a more detailed breakdown of MPC implementation effort of their study. Since then, as highlighted by Khabbazi et al. [7], more studies, including [9,18,19], have documented implementation efforts in terms of setup time and associated cost, such as labor and sensor upgrades. These studies offer valuable insights for estimating overall MPC cost effectiveness. However, many research studies treat MPC deployments as one-off demonstrations, making it difficult to gauge how much effort is reusable across multiple sites, such as software, modeling, and troubleshooting experience. These considerations are particularly important when considering scalability, as companies attempt to replicate MPC solutions across projects. To this end, woo Ham et al. [18,19] demonstrated a reusable solution for small commercial buildings with packaged rooftop units, successfully deploying MPC across 19 buildings with 56 Rooftop Units (RTUs). Despite this progress for small commercial buildings, similar outcomes have yet to be demonstrated for larger commercial buildings. Overall, the following challenges related to analyzing the benefits and costs for MPC, emerge from the literature:

1. There remains a lack of real demonstrations showcasing the value of price-responsive load shifting with MPC in large commercial buildings. This gap persists despite the increasing interest in dynamic pricing and demand response utility programs, as well as the significant potential impact of large commercial buildings on the total building stock.
2. The cost associated with implementing MPC remains uncertain, as the process requires specialized expertise for setup and maintenance. This uncertainty is particularly noticeable in large commercial buildings, where the variability in complexity and heterogeneity of their built-up HVAC systems warrant more customized MPC solutions for each building.
3. The value derived from MPC's ability to provide demand flexibility is contingent on the available controllable assets of the building (i.e. flexibility potential) as well as the economic incentives for flexibility (i.e. flexibility incentives).

1.3. Paper objectives and contributions

To address the challenges outlined in the previous section, this study focuses on implementing and testing price-responsive MPC in a real large commercial building HVAC system. The specific objectives of this paper are as follows:

1. Addressing challenge 1, extend testing of an MPC implemented in a real office building at Lawrence Berkeley National Laboratory (LBNL), initially for energy minimization [16], for price-responsive cost minimization. The MPC implementation was extended for price-responsive cost minimization in Zanetti et al. [29], where tests were carried out that demonstrated MPC response to two different price signal profiles in Summer and Fall seasons. In this regard, contributions of the present work compared to previous work are that two additional tests were carried out with two new price profiles in Winter and Spring, estimations are made of annual cost savings given data from all seasons, and demand shifting metrics are calculated for all seasons.
2. Addressing challenge 2, present practical challenges that were encountered and overcome when extending the MPC from the original energy minimization implementation in Blum et al. [16] to the price-responsive implementation in the current study and quantify the effort to make such an extension.
3. Addressing challenge 3, put cost savings estimates from this study in context with other experimental demonstrations from literature to trend estimates based on economic incentives for flexibility, in order to inform potential implementation cost targets for economic payback.

The paper is organized as follows. Section 2 describes the field demonstration site for the MPC implementation. Section 3 summarizes the MPC implementation and describes the updates done since [16]. Section 4 shows the MPC results in terms of cost and energy savings, flexibility provided, thermal comfort, and detailed performance analysis. Section 5 presents cost savings across experimental studies, discusses effort and practical challenges to update and deploy the price-based MPC, and discusses limitations and future work. Lastly Section 6 summarizes the conclusions.

Table 1
Summary of studies on field demonstrations of MPC in real commercial buildings or test chambers reporting cost savings, some of which provided some kind of demand flexibility or pricing scenario: VAV for variable air volume, UFAD for underfloor air distribution, RTU for roof top unit, AHU for air handling unit, TABS for thermally activated building structures, FCU for fan coil unit, HP for heat pump, PV for photovoltaic system, RC for resistance capacitance, ARX for Auto-regressive with exogenous input, MILP Mixed Integer Linear Programming.

Study	Year	Building	System	Control variable	Control objectives	Flexibility incentives	MPC approach	Test days	Results
[20]	2011	Office, USA, Illinois Lemont	VAV and chillers	AHU supply temperature set points	Minimize cost	Constant energy prices	Discrete transfer function for thermal envelope; quadratic program for optimal control	14	Energy savings between 6 and 45 % depending on test, cost savings extrapolated 5.9 $ $/m^2 y$
[21]	2014	Office, USA, Illinois Champaign	VAV	Flow rate and temperature of the mixed-air, discharge air temperatures of heating and cooling coil, and supply temperatures for each zone	Minimize cost	Constant energy prices	Semi-empirical models for thermal envelope and HVAC system; nonlinear program for optimal control solved by IPOPT	5	20 % cost saving for transition season, 70 % for heating season
[22]	2014	Offices in Newcastle and Melbourne, Australia	UFAD	Zone temperature set-points where available or supply air temperature if not	Minimize cost and CO_2 emissions	Constant energy prices	Discrete transfer function for thermal envelope; quadratic program for optimal control	60	19 % and 32 % for each building in the heating season
[23]	2015	Gym, USA, Philadelphia	Four RTUs	On/off status of RTU	Minimize energy and peak demand	No cost in objective function	ARX linearized to state-space; Linear program for optimal control solved by MATLAB	4	8 % building energy saving and 40 % peak demand reduction
[24]	2015	Airport terminal, Australia, Adelaide	Three chillers with AHU	Cooling valves and outdoor air damper position	Minimize cost and peak demand	TOU energy prices	Linear program using RC model for thermal envelope with ideal cooling heat rate as input and inverse neural network to convert optimal heat flow rate into valve and damper positions	3	13 % cost savings
[17]	2015	Office, Switzerland, Allschwil	TABS	Heating and cooling heat flux of TABS and AHU, air flow rate of AHU	Minimize cost	TOU energy prices	RC model for thermal envelope; bilinear problem for optimal control solved by CPLEX; thermal comfort as objective penalty	500	17 % HVAC energy saving and 5000 CHF/y in cost savings
[25]	2016	Office, Belgium, Brussels	AHU + FCU + radiator	Heating flux of the gas boiler and heat pumps	Minimize cost	TOU energy prices	Single zone RC model for thermal envelope implemented in Modelica; nonlinear program for optimal control formed using Jmodelica.org and solved by IPOPT; thermal comfort as objective penalty	90	34–40 % daily HVAC cost saving
[13,14]	2016	FlexLab, USA,California Berkeley	Chiller + hot water plant + VAV	VAV supply air temperature and fan speed	Minimize energy, estimate reserve capacity	Regulation and reserve signals	Linear and non linear programs using state space representation	4	8 % Mean Absolute Percentage Error with respect to the reserve capacity
[26]	2018	Retail store in, USA, Florida	Four RTUs	On/off status of RTU	Minimize energy and peak demand	No cost in objective function	RC model for thermal envelope with lumped disturbance model; integer linear program for optimal control; thermal comfort as objective penalty	30	12 % HVAC energy savings and 18 % HVAC peak demand reduction
[8]	2019	School, Italy	PV and battery	Battery power	Minimize cost	TOU energy prices with import-export differential	RC model for thermal envelope with lumped disturbance model; integer linear program for optimal control; thermal comfort as objective penalty	14	4 % to 7 % cost savings

Table 1
Continued

Study	Year	Building	System	Control variable	Control objectives	Flexibility incentives	MPC approach	Test days	Results
[12]	2021	FlexLab, USA, CA Berkeley	Single-zone AHU, PV and Battery	AHU supply temperature and flow rate, battery status	Minimize cost and peak demand	TOU energy prices, dynamic energy prices, load shifting and shedding, and load tracking	Reinforcement learning based on Deep Deterministic Policy Gradient	7	34 % cost savings during cooling season and 16 % energy reduction during heating. 50 % demand shift from high to low pricesleading to 0.25 W/ft^2 demand intensity reduction.
[9]	2022	Gas Station, USA, CA Blue Lake Rancheria	Two RTUs, PV and Battery	Thermostat set points and battery power	Minimize cost	Seasonal TOU energy prices, peak demand charges, and dynamic prices	RC-network model implemented in Modelica and optimized using IPOPT through JModelica	330	12 % cost savings, 34 % peak demand reduction, 3 % cost savings with dynamic price, able to shift up and down 10 kW for 100 % required time, able to shed 3 kW for 83 % of tasked time, able to track load profile 56 % of time
[15]	2022	Office, Switzerland, Zurich	Ground Source Heat Pump + Storage	heat pump load	Minimize cost and track flexibility services	Day-ahead TSO reserve market	Mix of state space and neural network to model building demand resulting in MILP solved with YAMILP	3	14 % electrical energy shift
[27]	2022	Office, USA, California	RTUs	Thermostat set points	Minimize cost	TOU energy prices	State space representation. Hierarchical MPC, first layer non linear optimization to find thermostat set points. Bottom layer peak power minimization and set point tracking	30	30 % cost savings and 40 % peak reduction
[11]	2023	Office, Singapore	Chillers with VAV and fancoils	Supply air flow rates to rooms	Maximize PV self-consumption	None	RC-network model implemented in Modelica and optimized using IPOPT through Jmodelica	49	PV self-consumption and building self-sufficiency were respectively 19.5 % and 10.6 %
[18]	2023	School, USA, California	RTUs	Thermostat set points	Minimize cost	TOU energy prices	State space representation. Hierarchical MPC, first layer non linear optimization to find thermostat set points. Bottom layer peak power minimization and set point tracking	30	16 % demand shift and 24 % peak reduction
[10]	2023	University, Switzerland, Lausanne	Heat Pump and Battery	Heat Pump load	Minimize cost	TOU energy prices, and variable reward for demand response	Hierarchical MPC with day ahead planner, online optimization and real time power adjustment. Data-enabled Predictive Control based on Willems' Fundamental Lemma that results in convex optimization problem	57	28 % cost savings
[19]	2024	Small commercial, USA, New York	Split heat pumps and gas furnace	Heat Pump load	Minimize cost	TOU energy prices	MPC based on grey box models and MILP problem formulation	13	27 % cost savings, 14 % peak demand reduction, 23 % heating load shift in the morning
[28]	2024	School, Canada, Montreal	Air source and ground source heat pump	Thermostat set points	Minimize cost	TOU energy prices	MPC based on greybox models to find optimal daily set points clustered by weather	1	44 % cost savings, between 47 and 95 % load shifting, 32 W/m^2 demand shift with air based systems, and 65 32 W/m^2 with radiant systems

2. Site description

This site description is based on that which is presented in Blum et al. [16], though summarized here for convenience and updated to reflect changes in baseline HVAC control since that study.

2.1. Building

Building 59 was built in 2015 and is located in the Lawrence Berkeley National Laboratory (LBNL) campus in Berkeley, CA, US. Berkeley enjoys mild climate, falling under ASHRAE Climate Zone 3C. The building has four floors, where the first floor is a mechanical room used to cool the high performance computing center located on the second floor. The MPC is applied to the HVAC system that explicitly serves the office areas on the third and fourth floors, encompassing approximately 6038 m^2 (65,000 ft^2). The third floor predominantly comprises enclosed office areas, while the fourth floor is primarily open office space. The third and fourth floors are thermally separated from the computing center on the second floor by a layer of rigid insulation under the third floor concrete slab. The building structure is steel-framed with an exterior metal curtain wall system with integrated windows and foamed insulation core. In office areas, finished floors with carpeting are raised above structural concrete slabs, creating the plenum for an Underfloor Air Distribution (UFAD) HVAC system. In terms of energy use of the office areas, the study in Luo et al. [30] analyzed data from 2018 to 2020 and found an Energy Use Intensity (EUI) of 104 kWh/m^2/year (33 kBtu/ft^2/year), which is within the highest frequency bin for the state of California (95 to 126 kWh/m^2/year or 30 to 40 kBtu/ft^2/year). Furthermore, the majority of Building 59 office area load is HVAC ranging from 60 % in 2019 to 88 % in 2020. While the very high percentage in the year 2020 can be attributed to COVID19 and very little occupancy, the data still shows that a relatively high percentage of the total building load is attributable to the HVAC system and controllable by the MPC.

2.2. HVAC and existing control

Heating, cooling, and ventilation are provided to the offices by a UFAD. The system uses four large rooftop units (RTUs) located on the roof with direct expansion (DX) coils to supply cool air to the common underfloor plenum on each level. Air diffuses directly to the core and perimeter zones through floor diffusers and can be pulled and heated to perimeter zones with fan-powered underfloor terminal units (UFT). Hot water for the UFT heating coils is produced by a 120 kW water-source heat pump (WSHP) located on the mechanical level, in the first floor of the building, where evaporator water comes from the same water loop as the DX compressors. There are no heating coils in the RTUs. This design is shown in cross-section for one RTU in Fig. 1.

Each RTU serves the third and fourth floor offices between particular column lines, though the areas of service are not separated by internal wall partitions. The design air flow of each RTU is 9.44 m^3/s, with design minimum outdoor air flow rate of 2.36 m^3/s during occupied hours. For each RTU, the supply fan motors are 14.9 kW and the return fan motors are 5.6 kW, each equipped with variable frequency drives (VFD). The cooling capacity of each RTU is 104 kWth with two 9.6 kW motor R410A scroll compressors with variable speed control of each compressor from 10 % to 100 %. There are 50 fan-powered underfloor terminal units (UFTs) with water heating coils to provide heating and increased air circulation. The condenser water from the RTUs is cooled by cooling towers located next to the building on the mechanical level. These cooling towers are shared with the high performance computing cooling equipment, which dominates the load on the towers.

The HVAC system is controlled by an Automated Logic Corporation (ALC) WebCTRL Building Management System (BMS) [31]. The occupancy schedule for weekdays, from 5 a.m. to 10 p.m., determines the HVAC operation. RTU outside air dampers are controlled to meet at least the minimum outside airflow requirement during these times using outside air flow measurement stations, and allowed to close when the building is unoccupied. The perimeter zone thermostat heating and cooling set points are based on this occupancy schedule, as shown in Fig. 2. Set point changes before 5 a.m. are caused by pre-cooling and pre-heating logic implemented in the baseline controller. Finally, the RTUs are turned off during unoccupied hours unless there is a heating or cooling request as determined by the unoccupied set points. Thermostats in the core zones were added during testing and were not used in the MPC, instead 16 Raspberry Pis [32] equipped with a temperature sensor are located in the core zones across the two floors.

Perimeter zone thermostats control the UFTs fan speed based on cooling and heating demand, as well as the hot water valve position for heating. However, to keep the noise of the UFT fans low for occupants and as a result of system commissioning, strict limits are placed on some UFT fan speeds typically ranging from a minimum of 20 % and maximum of 50 %. It is important to note that thermostats in the core zones of the building only trigger heating and cooling requests and do not control UFTs, as supply air diffuses directly from the underfloor plenums into core spaces. All four RTU supply fan speeds are controlled by a single PI controller tracking a variable static pressure set point for the average underfloor plenum static pressure measurement of all units. The pressure set point is between 3.75 and 12.5 Pa and is calculated through a trim and respond strategy depending on the number of zone cooling requests, starting from the minimum value and increase as cooling requests increase. The return fan speeds of each RTU are controlled to track a flow rate equal to 95 % of the supply air flow rate minus an additional 0.1 m^3/s. The supply air temperature for each RTU is controlled to a variable set point, between 14.4 and 22.2 °C that depends on the difference between the number of zone heating and cooling requests, and it is calculated through a trim and respond strategy by raising the set point as the difference increases. The supply air temperature set point is maintained by the DX coil and/or economizer depending on outside air dry bulb temperature conditions.

3. MPC implementation

This section briefly summarizes the MPC implementation with a focus on updates from the study from Blum et al. [16]. For a more in depth explanation on the MPC implementation please refer to Blum et al. [16].

3.1. Software architecture, data exchange, and BMS integration

The MPC software architecture consists of eight main services that run concurrently, presented in Fig. 3. Each service runs within its own Docker [33] container, though each container is based on the same Docker image. These services run on a dedicated server on campus and exchange data with various Building 59 and site data collection systems, web-based services, and the project database. This database is an instance of an InfluxDB timeseries database [34]. The *Control Optimization*, *State Estimation*, and *Parameter Estimation* services are implemented using the open-source software MPCPy [35] commit `ecb9833`. This workflow utilizes a user-defined model written in Modelica [36] along with user-defined data specified in a Python environment to automatically formulate and solve the appropriate optimization problems for each of the control optimization, state estimation, and parameter estimation services.

Fig. 3 also shows the different systems and methods to gather data. Separate Python modules were developed to collect data from each building system, since each building system requires use of a separate API or communication protocol. The *Historic Data Collection* service runs on a single Python module that centralizes the use of each of the system-specific modules to collect all system data. The MPC solution is calculated by the *Control Optimization* service and stored in the database every 10 min. Every one minute, the most recent valid solution set points are sent to the building Building Management System (BMS) via the *Send*

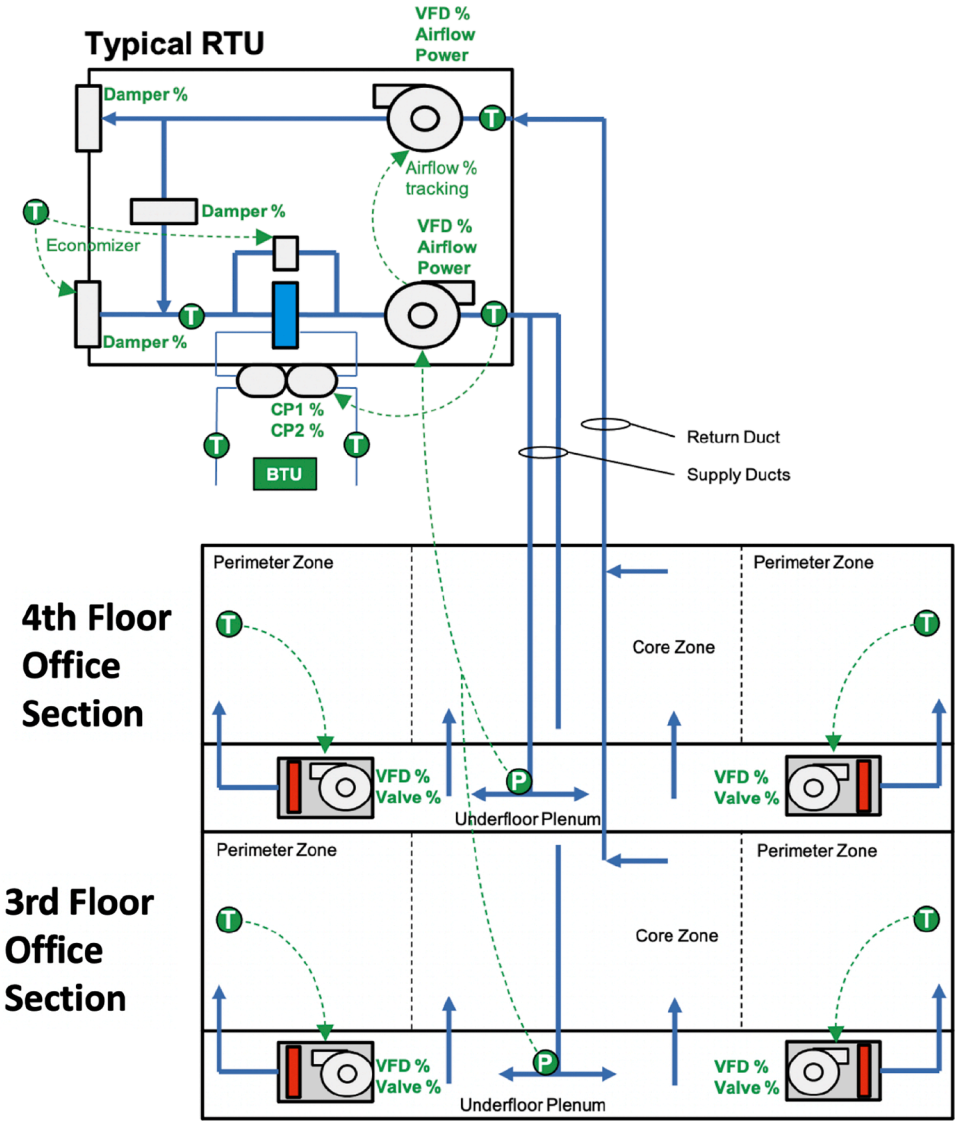


Fig. 1. Design and control schematic in section view for one RTU of the UFAD HVAC system. Important sensor and control points are labeled, including temperatures (T) and static pressures (P) [16].

Set Points service. The integration logic between these set points and the ALC BMS is shown in Fig. 3.

Highlighted in red in Fig. 3 are the updates with respect to the previous study [16]. In particular:

- The electrical plug load data previously obtained via an Elastic Search (ES) database Query are now obtained via a campus SkySpark API [37].
- A service to retrieve day-ahead electricity prices was implemented to obtain data from the CalFlexHub and MIDAS servers via HTTP [38]. The electricity prices are used in the MPC objective function to estimate the energy cost over the prediction horizon.
- The Weather Forecast service was changed from using DarkSky [39] to the National Oceanic and Atmospheric Administration (NOAA) API [40].
- Since the study in Blum et al. [16] was carried out, the building facility managers implemented schedules that reset zone temperature set points and outside air requirements based on the occupancy schedule. Therefore, an additional service was added to the MPC to retrieve historical temperature set points, outside air requirements, and occupancy schedule, and use them to define constraints in the MPC formulation.
- During the same control update, the facility managers implemented scheduling that turned off the HVAC system during unoccupied hours. This increased the MPC control space, allowing it to turn off the RTUs during unoccupied hours and turn them back on to preheat or pre-cool the building. For this reason, an additional set point, *unocc_run*, was added to the list of points controlled by MPC. The facility staff implemented this new set point in ALC.
- The offset set point control input *rtui_znt_adj_stp* available for the thermostats in the perimeter zones was repurposed to initiate heating mode. The value is set to the difference between the MPC forecasted temperature and current set point. If this difference is positive it will raise the thermostat set point and potentially trigger heating requests that would turn on the UFTs and WSHP. If the difference is negative the input is ignored since cooling is controlled directly via the RTUs. Lastly the thermostats inputs are grouped based on the same zone division principle used for the RTUs.

3.2. Modeling and parameter estimation

The models described here are used by the MPC controller in real time to predict the performance of the building and optimize control.

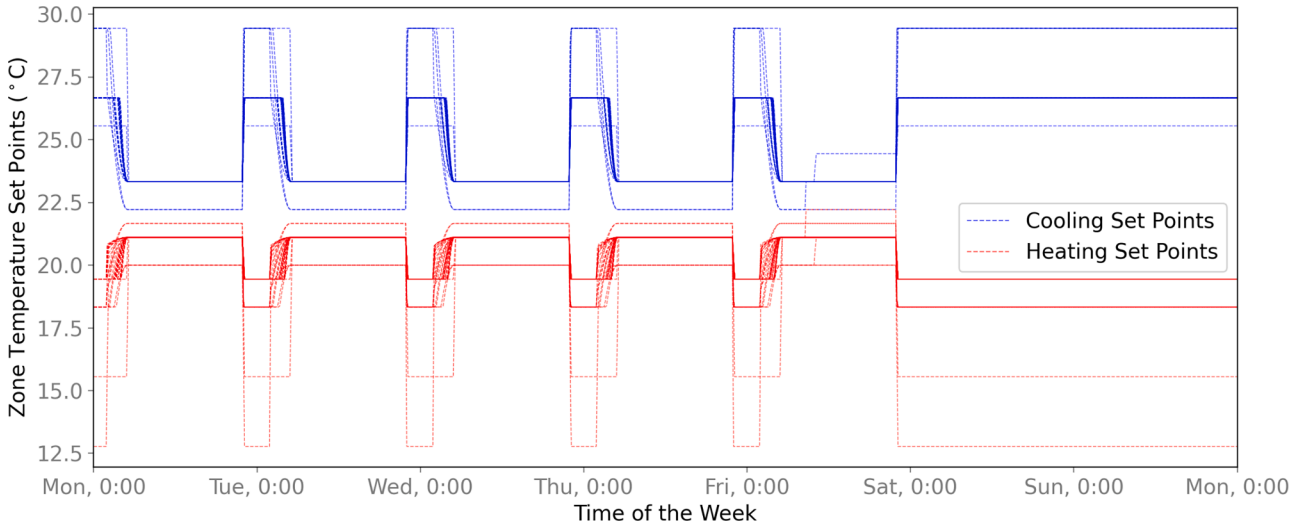


Fig. 2. Typical weekly schedule for all the set points in the perimeter zones that have a thermostat. This weekly profile repeats for the whole year, in blue are reported the cooling set points and in red the heating set points. Occupancy is considered from 5 a.m. to 10 p.m. on weekdays. Set point changes before 5 a.m. are due to pre-cooling and pre-heating logic implemented in the baseline controller.

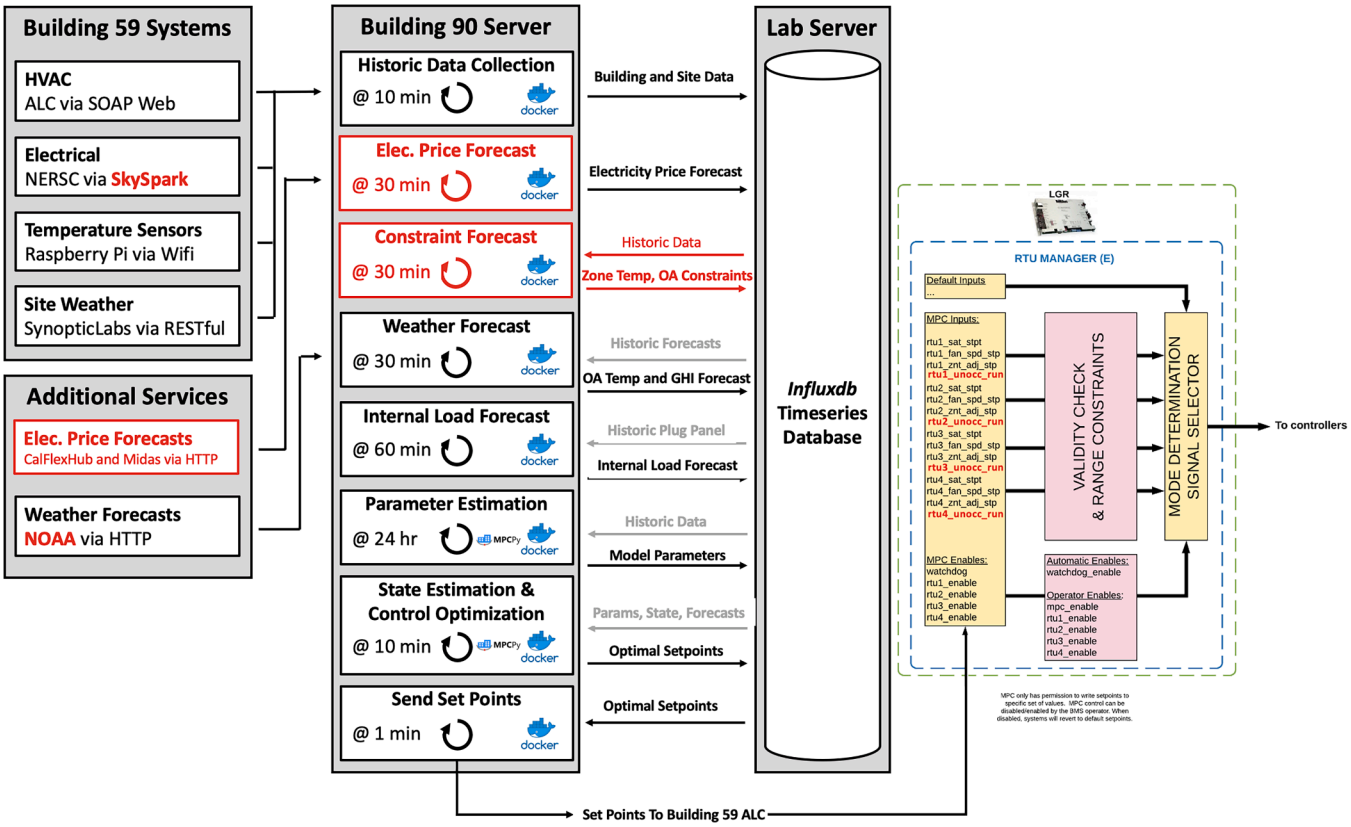


Fig. 3. Schematic of the MPC software architecture, data integration, and building BMS communication, with updates from the previous study highlighted in red.

Important values to predict are building space temperatures, RTU air flow rates and temperatures, fan power consumption, DX compressor power consumption, and UFT heating demand. All the models used for the MPC can be considered grey-box, utilizing simplified, yet physically-based, models implemented in Modelica for use with MPCPy. In particular, an equivalent resistance capacitance circuit (R2C2) was used to model each of four thermal zones, one for each RTU. The choice behind and verification of an R2C2 circuit is provided in more detail in the previous study [16], and stems from earlier work in Blum et al. [41], where an extensive sensitivity analysis on the effect of model structure on MPC performance demonstrated a trade off between simplicity, benefiting robust parameter estimation, and complexity, benefiting representation of proper dynamics. Field data was used to estimate the parameters of each model. Each RTU zone corresponds to a collection of several perimeter and core zones attributed to being served by the RTU. The single zone temperature for each RTU is estimated by weighting all the

corresponding zone temperatures by their design UFT airflow rate. The RTU model contains a mixing box, supply and return fans, and DX models. The mixing box mixes outside air and return air in a ratio specified by the outside air fraction control and supply air massflow rate signals. The supply fan takes a specified flow rate set point. As a model simplification, the return fan is assumed to move an equal amount of air. The power consumption of the fan is estimated as a cubic function of the air massflow rate. The DX is modeled as a simple heat exchanger removing sensible heat from the air stream. Its power consumption is estimated as a quadratic function of the heat flow rate.

The model parameters for the RC network and fan and DX power consumption curves are considered time-invariant for each iteration of MPC optimization. Due to the simplification described above, real time-invariant parameters may not be able to properly characterize the behavior of the real system in all its operating points. To account for this mismatch, as well as other seasonal variations in performance, the model parameters for envelope and fan are updated daily. The new parameters are calculated with an optimization problem that minimizes the error between measurement and model prediction for the most recent data, split between training and validation periods, using up to a 20-day time period for the envelope, with minimum of one week of consecutive days depending on data quality, and 15-day time period for the fan, with a minimum of three consecutive days. Because there are no direct measurements of the DX coil and the training process was more tailored, these parameters were determined only once with operational data from August 1, 2020 to October 18, 2020, as reported in Blum et al. [16].

3.3. Forecasts

Weather forecasts are collected from NOAA [40] and include drybulb temperature, humidity, dew point, pressure, wind speed, wind direction, and cloud cover. The global horizontal irradiation (GHI) is not provided by the service and it is instead estimated from cloud cover data, drybulb temperature, and clear-sky irradiation. This estimation is obtained using a data-driven model trained on historical data obtained from a weather station situated on LBNL's campus. The forecast data is provided with hourly time steps for a 24-h period and updated approximately every 6 h. See [16] for details. The dynamic electricity price is collected from the CalFlexHub [42] and MIDAS [43] servers. It consists of a 24-h profile with a 15 min time step. There are four typical profiles, one for each season. These are synthetic profiles adapted from those defined by Gerke et al. [44], which use an approach similar to the CalFUSE program [45] to study demand response potential across California. The four profiles used in this study are shown in blue in Fig. 5 in the next section.

Internal loads consist of various internal heat gain sources present in the building such as plug load, lighting, and occupant heat gains. Plug load measured at plug electrical panels was used as a proxy of total internal loads and a data-driven model is used to predict the plug load based on the hour of the day and day of week. The model uses the average value of the same hour of day and the same day of week of the previous 90 days. Temperature set point forecasts, used to define operational constraints in the MPC control problem formulation, are based on the weekly schedule shown in Fig. 2 established by the building operators. Temperature set points correspond to cooling and heating set points, creating a comfort band. Each zone with a thermostat is assigned a weekly schedule and from there a weighted average is obtained for each of the aggregated zones used by the MPC. The other occupancy-based set point is the minimum outside airflow requirement in each RTU.

3.4. Control optimization and state estimation

This section provides only a summary of the optimal control problem formulation and the extension compared to the original implementation [16]. The optimal control problem is defined over a 24-h horizon and solved separately for each RTU and defined for the current time, t_0, to the time at the end of the prediction horizon, t_f, by

$$\min_{u_{sf},u_{dx},u_{oa},u_{hw},s_1,s_2,s_3} J = \int_{t_0}^{t_f} (w_0 p_{ele} P + w_1 s_1^2 + w_2 s_2^2 + w_3 s_3^2 + k_1 u_{sf}^2 + k_2 u_{dx}^2 + k_3 u_{oa}^2 + k_4 u_{hw}^2) dt \quad (1)$$

$$\text{s.t.} \quad f(\dot{\mathbf{x}}, \mathbf{x}, \mathbf{u}, \tilde{\boldsymbol{\omega}}, \boldsymbol{\theta}_e^*, \boldsymbol{\theta}_f) = 0 \quad (2)$$

$$\mathbf{x}(t_0) = \mathbf{x}_0^* \quad (3)$$

$$T_z - s_1 \le T_z^u \quad (4)$$

$$T_z + s_2 \ge T_z^l \quad (5)$$

$$T_{sa} + s_3 \le T_{sa}^l \quad (6)$$

$$T_{sa} \le T_{sa}^u \quad (7)$$

$$\dot{m}_{sa} \ge \dot{m}_{sa}^l \quad (8)$$

$$\dot{m}_{oa} \ge \dot{m}_{oa}^l \quad (9)$$

$$0 \le u_{sf}, u_{dx}, u_{oa}, u_{hw} \le 1 \quad (10)$$

$$s_1, s_2, s_3 \ge 0. \quad (11)$$

The total power P is equal to the sum of supply and return fan power, DX power, and UFT reheat power, p_{ele} is the dynamic electricity price, making this a cost minimization objective and w_0 is the associated weight. u_i correspond to the MPC control variables, where u_{dx} is the DX control signal, u_{oa} is the outside air damper control signal, u_{hw} is the UFT hot water valve control signal, and u_{sf} is the supply fan control signal. Upper and lower constraints on zone temperature, T_z^u and T_z^l are time-varying according to the occupancy-based set points described in the previous section and a lower constraint on supply air temperature, $T_{sa}^l = 17.2\,°\text{C}$, are implemented with slack variables s_1,s_2,s_3 and corresponding weights w_1,w_2,w_3. These constraints are implemented with slack variables instead of hard constraints to ensure feasibility. This is typical for zone temperature and used for the lower limit of supply air temperature in the case the outside air temperature is too low to maintain the minimum supply air temperature and minimum outside air flow requirement. The upper limit on supply air temperature is constant, $T_{sa}^u = 21.1\,°\text{C}$, while the minimum supply air flow and minimum outside air flow are time-varying, where $\dot{m}_{sa}^l = 4.75\ \text{m}^3/\text{s}$ and $\dot{m}_{oa}^l = 2.36\ \text{m}^3/\text{s}$ during occupied hours and zero otherwise. These three constraints are implemented as hard constraints, along with control signals to be between 0 and 1 and slack variables to be non-negative. These numerical constraints were added to ensure operation of the HVAC system under MPC is safe and consistent with prevailing preferences by facility operators. For example the minimum supply air flow constraint is added as a safety measure to ensure sufficient air flow is provided through the DX coil when RTU is determined on to prevent freezing, the minimum outside air flowrate during occupied times is consistent with ventilation requirements, and the upper limit of the supply air temperature is consistent with typical system operation. Regularization on the RTU control signals is added in the objective function for the opportunity to help prevent oscillatory control solutions with weights k_1,k_2,k_3,k_4. For this field study, through offline tuning, it was determined that all weights be equal to 1 except $k_1 = 1e4$, while during online operation the cost weight was tuned to a final value of $w_0 = 5e5$. In addition, $\boldsymbol{\theta}_e^*$ is the set of estimated parameters resulting from the parameter estimation processes, $\tilde{\boldsymbol{\omega}}$ are forecasted disturbances, and $\mathbf{x}_0^*$ is the initial state estimation by the state estimator. The state estimation consists of a simple moving horizon estimator. The estimator solves an optimization problem from the start of historic time horizon (previous 24 h) to the current time, where the objective is minimizing the error between measured and estimated zone temperature and the optimization variables are the state temperatures at the initial time. Then, the state temperatures at current time are used to initialize the optimization at the current time step.

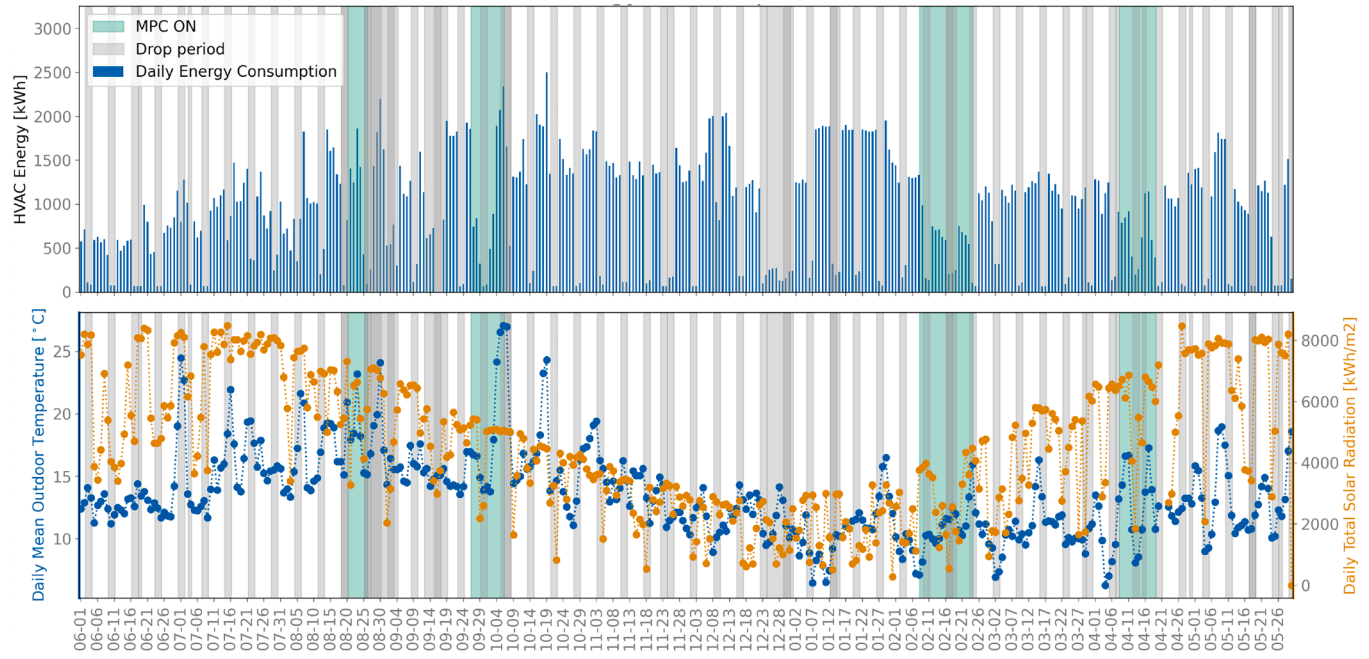


Fig. 4. HVAC daily energy use (top) and daily average external dry bulb temperature and daily horizontal radiation (bottom). Highlighted in green are the MPC testing periods and in grey are the days that were removed from the test.

4. MPC performance and analysis

This section presents the results obtained during the tests performed in Summer 2023, Fall 2023, Winter 2024 and Spring 2024. Fig. 4 shows the overall period under analysis (6/1/2023–5/30/2024) and the MPC testing periods (8/21/23–8/26/23, 9/27/23–10/6/23, 2/9/24–2/24/24, and 4/9/24–4/19/24) highlighted in green. These four periods allow us to extrapolate the MPC performance for the whole year to estimate yearly savings for the MPC. The grey shaded areas are periods not considered in the results analysis. This is to avoid weekends and holidays where the HVAC was mostly off, and periods where HVAC operation was abnormal due to faults or data failures that inhibited the ability for the MPC to operate continuously.

4.1. Energy use and cost

Hourly energy use distributions for each season's testing period for when MPC was active (MPC ON) and when baseline control was used (MPC OFF) are shown in Fig. 5. The hourly energy use E_h is calculated by using resampled power measurements from two HVAC electrical panels P_{hvac1} and P_{hvac2}, the sum of all four RTUs supply and return fan power measurements p_{fantot} reported by the individual fan Variable Frequency Drives (VFD), the measured reheat water heat flow rate at the WSHP, Q_{wshp}, and estimated WSHP COP, COP_{wshp}, assumed to be 3.0 based on the specification sheet. This estimation is due to no measurements being available of the electrical draw of the WSHP. The resulting energy use E_h is calculated according to

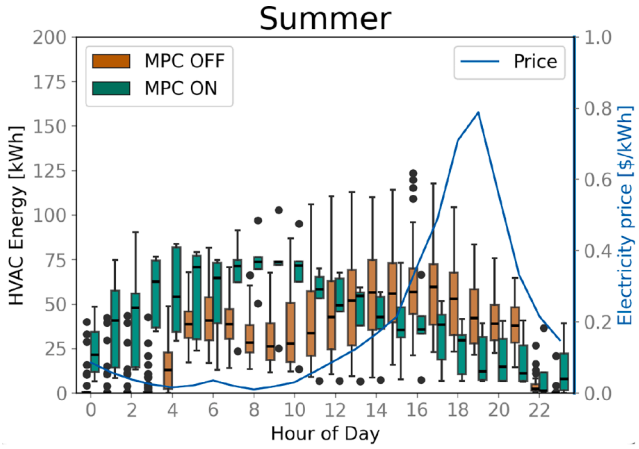
$$E_h = max(P_{hvac1} + P_{hvac2}, p_{fantot}) + \frac{Q_{wshp}}{COP_{wshp}}. \quad (12)$$

The maximum between the sum of the measured HVAC panel powers and the total fan power reported by the VFDs is used because the panel measurements are reported by a smart breaker that has a threshold for the minimum power that it can read, below which it reports zero. Therefore, when the RTUs consumption is low, meaning no compressors are online and the fans are running at low speed, the breaker sometimes reads zero power, and the fan VFD measurements should be used instead to report power. In the Appendix, Fig. A.13 verifies that using VFD measurements can represent HVAC panel power at very low loads.

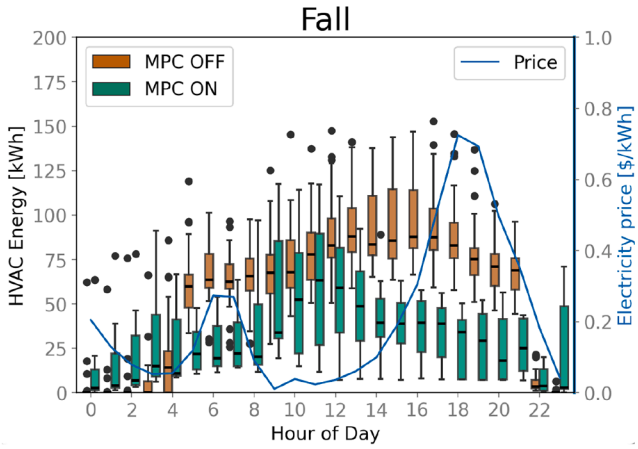
Fig. 5 shows that the MPC is able to shift HVAC load in response to dynamic electricity prices. The spread in the hourly energy distributions indicates the difference in outdoor temperature, radiation, and internal gains for each day under analysis. In the Summer, the MPC is able to shift the demand away from the high-price period in the afternoon thanks to pre-cooling and free cooling of the building at night and early morning times. In the Winter, the MPC is able to shift the demand from the afternoon peak and reduce energy use by pre-heating the building and then keeping a lower supply air temperature that allows the fans to run at reduced speed while maintaining the same outdoor air requirements. In the Spring and Fall, the situation is more mixed. If a day is colder, then it is similar to a winter scenario, while if it is warmer, it is more similar to a Summer scenario. Furthermore, in Spring and Fall, the MPC seems to shift HVAC usage to the midday to avoid both the morning and evening price peaks. More detailed performance of the MPC is presented in Section 4.3. Fig. 6 shows the daily energy use and cost plotted against mean outside air temperature for MPC ON and MPC OFF, along with piece-wise linear regressions generated using the inverse modeling open-source software described in Li et al. [46]. The figure shows that the MPC not only was able to reduce energy costs thanks to shifting energy to avoid high cost periods, but it did so while also reducing energy use, primarily in the Fall, Winter, and Spring seasons.

4.2. Annual energy and cost savings with price sensitivity

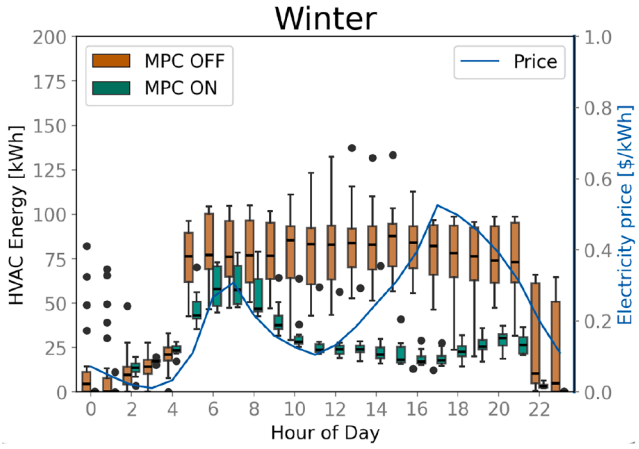
In order to extract the expected typical annual energy and cost savings, the piece-wise linear (univariate change-point) models, f_{cp}, shown in Fig. 6 are used similarly to the measurement and verification procedure recommended in ASHRAE Guideline 14–2023 [47]. The model is used to predict energy use, $\hat{E}_a$, (and similarly cost) for both MPC ON, $\hat{E}_{a,on}$, and MPC OFF, $\hat{E}_{a,off}$, from a single independent variable, the mean daily dry-bulb temperature $T_{mdb,d}$. Then, we use Typical Meteorological Year (TMY3) data for Oakland, CA for each day of the year, d, to extrapolate performance to a statistically typical year of weather



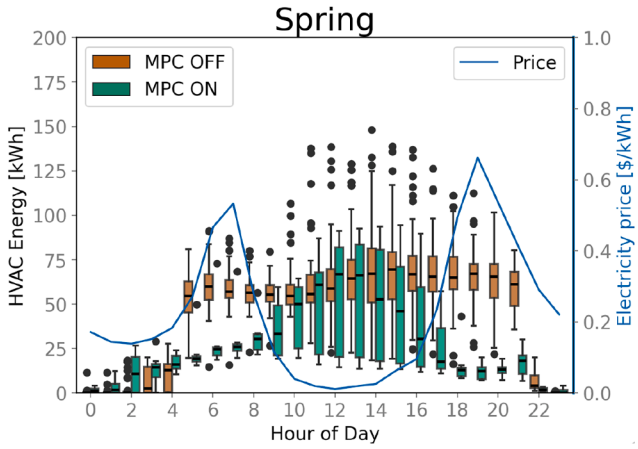
(a) Summer HVAC daily electricity consumption box plot



(b) Fall HVAC daily electricity consumption box plot



(c) Winter HVAC daily electricity consumption box plot



(d) Spring HVAC daily electricity consumption box plot

Fig. 5. Hourly distribution of HVAC energy use for each day of the analysis period, for (a) Summer (6/1/23–8/27/23), (b) Fall (9/4/23–11/30/23), (c) Winter (12/1/23–2/29/24), and (d) Spring (3/1/24–5/30/24) test periods, with MPC ON (green) and OFF (red). Excluded are days as shown in Fig. 4. In addition, on the right *y*-axis is shown the seasonal dynamic electricity price profiles, identical for each day of the season (blue).

data, according to:

$$\hat{E}_a = \sum_{d=1}^{365} f_{cp}(T_{mdb,d}). \tag{13}$$

Another locally measured climatic variable is solar radiation which could have been used in addition. However, inclusion of it only brought very marginal improvement to the model fit to the MPC OFF models, and no statistically significant improvement to MPC ON models at all. Due to the lack of substantial benefits, the univariate model formulation was used to eliminate multicolinearity and reduce data collection requirements, as supported by Kissock et al. [48], Fu et al. [49]. Annual energy savings, $\hat{E}_{a,sav}$, are then computed from

$$\hat{E}_{a,sav} = \hat{E}_{a,off} - \hat{E}_{a,on}. \tag{14}$$

Furthermore, the uncertainty in the calculation for annual energy use, $\Delta^2\hat{E}_a$, for each MPC ON, $\Delta^2\hat{E}_{a,on}$, and MPC OFF, $\Delta^2\hat{E}_{a,off}$, is calculated by Reddy et al. [50]

$$\Delta^2\hat{E}_a = 1.6 \cdot MSE \cdot n_{projection} \cdot \left(1 + \frac{2}{n_{training}}\right) \tag{15}$$

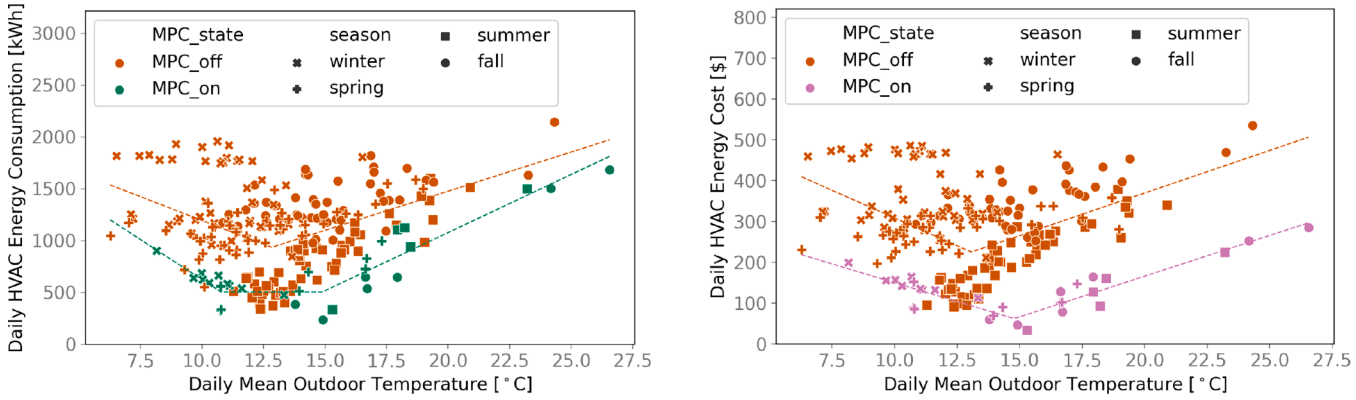
where MSE is the mean square error of the model, $n_{projection}$ is the size of the projection set, equal to 261 days, which include all weekdays excluding weekends, and $n_{training}$ is the size of the training set, equal to 30 days for MPC ON and 187 days for MPC OFF. Then, the uncertainty in the savings, $\Delta^2\hat{E}_{a,sav}$, is calculated by

$$\Delta^2\hat{E}_{a,sav} = \Delta^2\hat{E}_{a,off} - \Delta^2\hat{E}_{a,on}. \tag{16}$$

Annualized cost savings and uncertainties are computed similarly.

To perform a sensitivity analysis on annual cost savings around price profile, two new price scenarios were developed, shown in Fig. 7, and used to project annual cost savings according to the procedure described above. The *default* scenario was used during testing. Meanwhile, the *scaled* scenario represents one where the ratio between daily price peak and trough was reduced and the *constant* scenario represents a constant price throughout the day, but slightly variable across the seasons. The *scaled* price is obtained by solving an optimization problem aimed at reducing the ratio of peak to trough, while maintaining the same area under the curve and a similar curve shape as the *default* price. The method is explained in more detail in Appendix B. The *constant* price is obtained by taking the area under the curve of the *default* scenario and dividing it by 24 h. Figures for daily HVAC energy cost using test data similar to Fig. 6 for these two additional price scenarios can be found in the appendix as Fig. A.14.

Table 2 shows the model goodness-of-fit metric, CV-RMSE, and the estimated annual savings for energy use and cost for each price scenario. Although the CV-RMSE values of some of the regression models were above the recommended 22.5 % threshold in ASHRAE Guideline 14, the savings uncertainties are small. This is consistent with



(a) HVAC energy use scatter plot

(b) HVAC energy cost scatter plot with default prices

Fig. 6. Daily HVAC energy use (a) and cost (b) plotted against the daily mean outdoor temperature for the electricity prices shown in Fig. 5, with piece-wise linear regression (dashed line). Excluded are days as shown in Fig. 4.

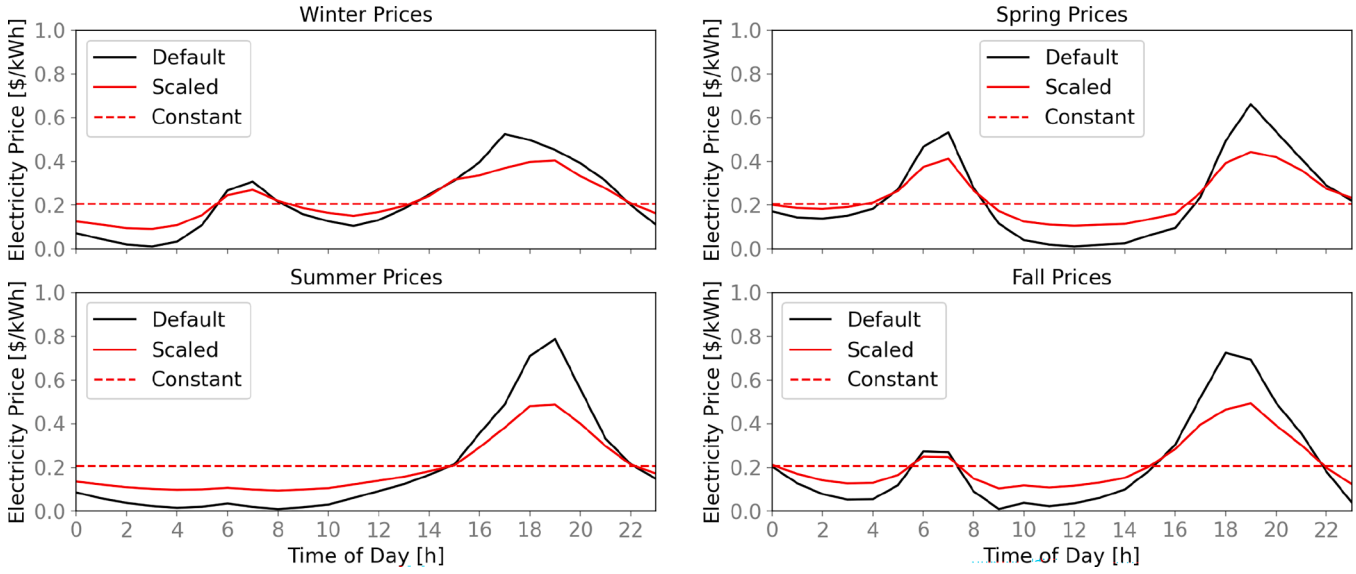


Fig. 7. Daily electricity price profiles for the different seasons. Default (black) used in the MPC objective, scaled (red), and constant (red dashed). Appendix B reports how scaled and constant profiles were obtained from the default profile.

Table 2
Model metrics and MPC savings annualized using the change point models in Figs. 6 and A.14 using TMY3 Oakland, CA weather file.

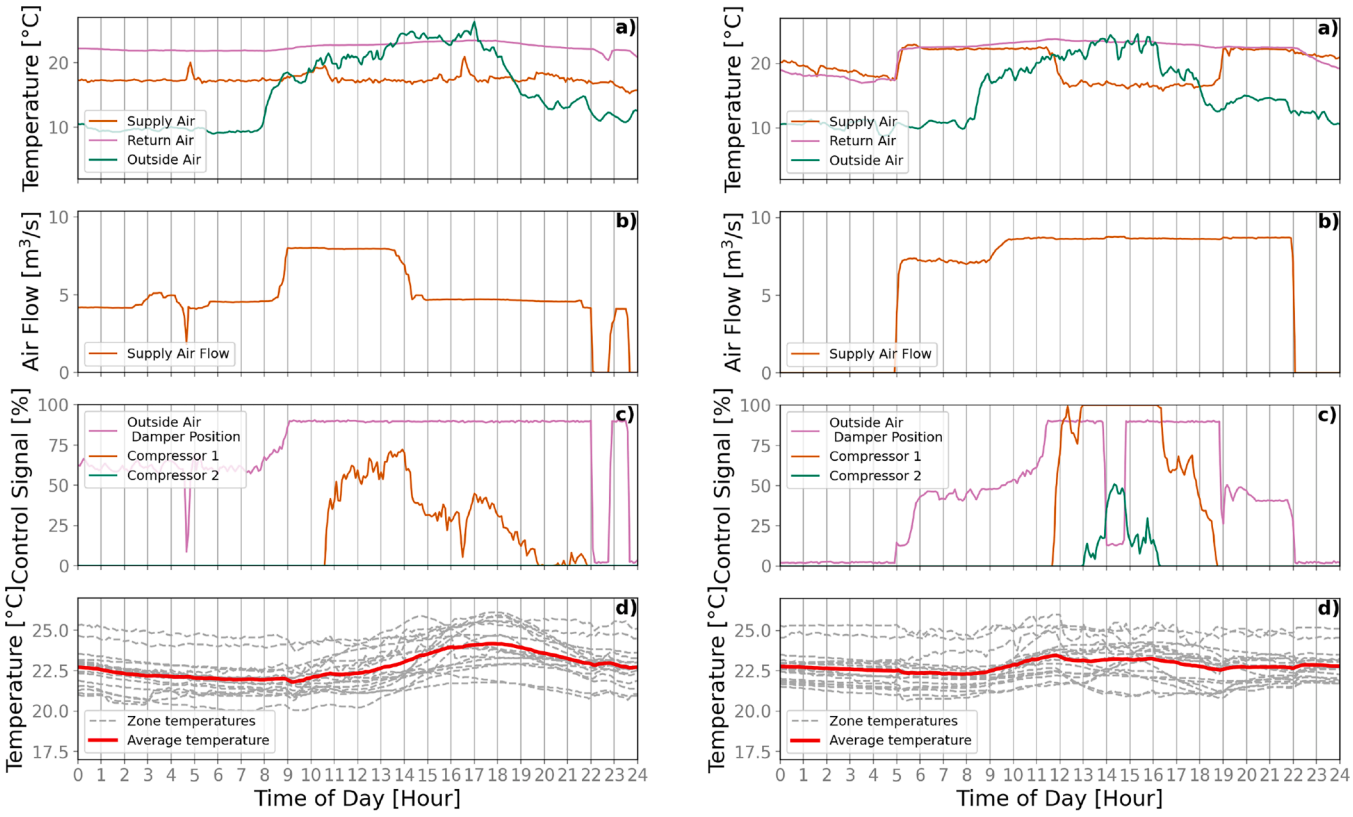
	CV-RMSE		Savings
	MPC OFF	MPC ON	
Energy	29.4 %	20.6 %	135,055 ± 8,873 kWh 45.2 % ± 5.7 % 22.4 ± 1.5 kWh/m^2
Cost (default)	30.2 %	18.5 %	45,492 ± 1,919 USD 61.5 % ± 4.2 % 7.5 ± 0.3 USD/m^2
Cost (Scaled)	30.3 %	17.1 %	38,280 ± 1,839 USD 54.5 % ± 4.8 % 6.3 ± 0.3 USD/m^2
Cost (Constant)	31.8 %	20.7 %	27,336 ± 1,730 USD 44.2 % ± 6.3 % 4.5 ± 0.3 USD/m^2

the examples given by Reddy and Claridge [51], where the authors demonstrated that when the savings itself was substantial enough, the uncertainties in models did not have much influence on the uncertainty of the savings.

4.3. Detailed performance analysis and thermal comfort

Figs. 8 and 9 show a representative day of operation for one of the RTUs (RTU1) during a warm and a cold day for the MPC ON and MPC OFF scenarios. During the warm day MPC OFF operation, in Fig. 8(2), the controller turns on the fan according to the occupancy schedule at 5 a.m. Then, it keeps nominal fan operation and mild supply air temperature throughout the start of the day. In the middle of the day as the number of zone cooling requests rises, the static pressure set point rises, increasing the fan speed, and the supply air temperature set point is lowered, thus turning on the DX compressors. Meanwhile, for MPC ON, shown in Fig. 8(1), the MPC turns the RTU on at night with low supply flow rates to pre-cool the building, taking advantage of the low outside air temperature, with minimal fan power consumption. This pre-cooling action allows the MPC to start lowering the supply fan speed and airflow around 2 p.m., which is when the electricity price starts to increase as shown in the Fall plot of Fig. 5, while keeping the average temperature of indoor air within a reasonable level.

During the cold day and MPC OFF operation, shown in Fig. 9(2), the baseline controller MPC OFF keeps a mild supply air temperature and high fan air flow rate throughout the occupied day. Meanwhile, during MPC ON operation, shown in Fig. 9(1), the MPC reduces fan



(1) Typical warm day (9/28/23) MPC ON

(2) Typical warm day (10/12/23) MPC OFF

Fig. 8. RTU1 example warm day plot. Plot a) shows Outside (green), Supply (orange) and Return (violet) air temperatures. Plot b) shows Supply (orange) air flow rate. Plot c) shows Compressor 1 (orange), Compressor 2 (green), and outside air damper (0 closed, 100 open) control signals. Plot d) shows Zone Temperatures (dashed grey) with Average (red). MPC ON in Figure (1) and MPC OFF in Figure (2). MPC ON is on 9/28/2023 and MPC OFF is on 10/12/2023.

energy throughout the day. However, it has to comply with the minimum supply air temperature set point and minimum outdoor air requirements, which causes the MPC to pre-heat the building during early morning around 4:30 a.m. by raising the set point in the zones and triggering UFT recirculation heating (without RTU airflow). Then, the RTU turns on at the start of occupancy and the fan speed is set to maintain the minimum outside air requirement and recirculate enough air to maintain the minimum supply air temperature. When the outdoor temperature rises and the constraint on supply air temperature relaxes the MPC lowers the fan speed as low as possible while maintaining the minimum outside air volume requirement.

Fig. 10 shows the distribution of temperatures for all zones of the building for each hour of the day with MPC ON and MPC OFF for each season. In the previous study [16], under baseline control, the RTUs ran 24/7 without turning off, and the baseline controller was able to keep a tighter temperature distribution with respect to the MPC. Instead, in the current study, the baseline controller operates on an occupancy schedule and static pressure and supply air temperature resets based on cooling/heating requests. Fig. 10 shows that in this study, the MPC is able to keep a tighter temperature bound than the baseline controller for the most part. However, there was slightly more temperature excursion in the MPC case in the late afternoon during the Fall and Summer tests, seen in Fig. 10 where the edge of the whiskers for MPC are above the baseline controller. Contributing to this observation is that the outside air temperature reached unusually high levels during Summer and Fall MPC tests, as can be observed in the bottom plot of Fig. 4. Under MPC control, the total air flow into the plenum reduces in the afternoon, thus lowering the air flow through the perimeter zone UFTs, coupled with the reduced speed at which UFTs can run to avoid excessive noise, and sun exposure of perimeter zones.

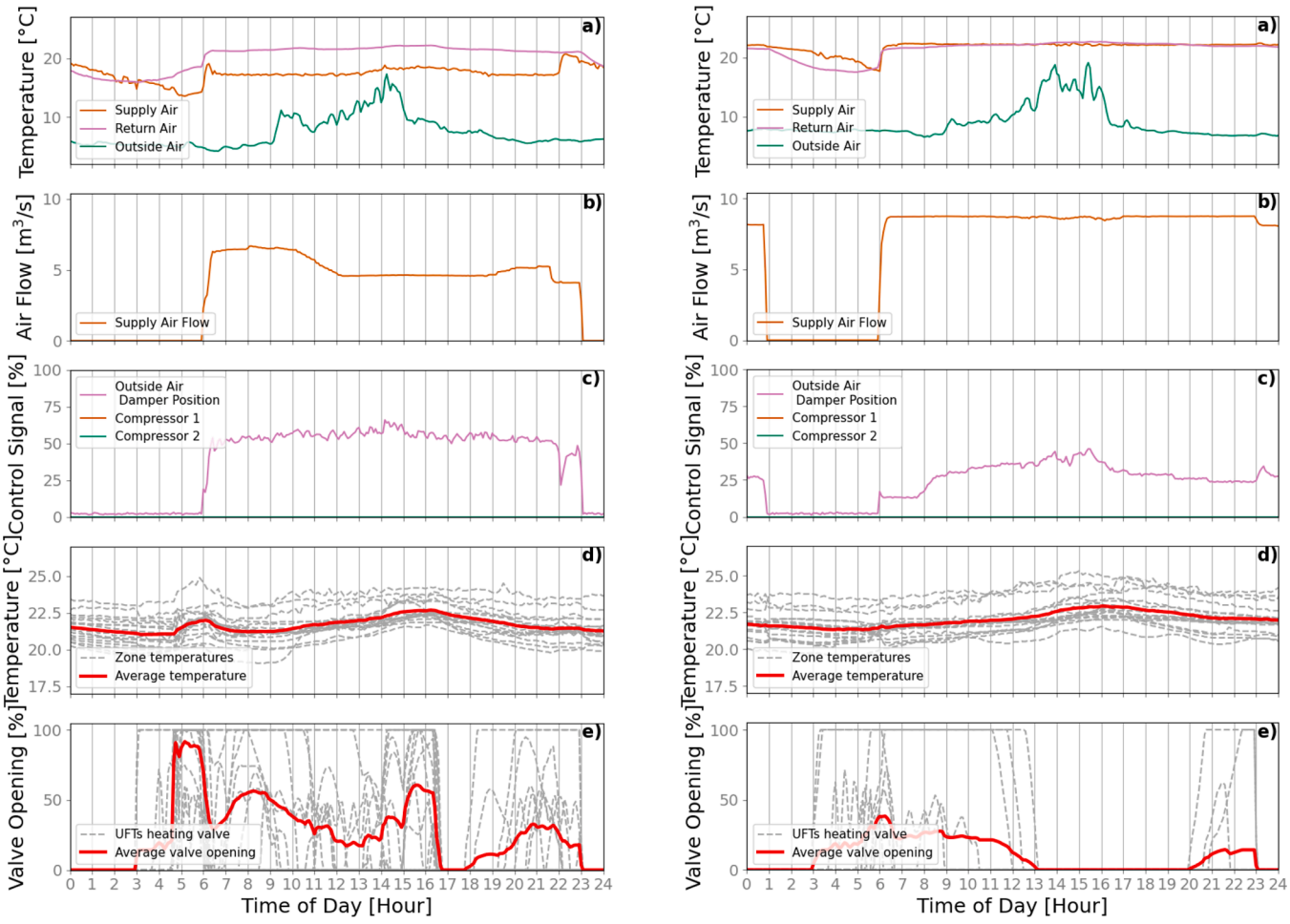
The zone-averaged daily discomfort for a day d, denoted as D_d, is calculated using the deviation between the zone temperature measurement, T_z, of each zone z for a total number of zones Z, and that zone's heating and cooling set points, ${T_z}^h$ and ${T_z}^c$ according to:

$$D_d = \frac{1}{Z}\sum_{z}^{Z}\int_{t_0}^{t_f} max({T_z}^h - T_z, 0) + max(T_z - {T_z}^c, 0)dt \tag{17}$$

The set points are shown in Fig. 2 for the perimeter zones, while for the core zones measured by the Raspberry Pi sensors the average of the surrounding perimeter zones was used instead. Fig. 11 shows discomfort data as a function of daily mean outside air temperature under MPC ON and MPC OFF. Dashed lines show piece-wise linear regressions for each of MPC ON and OFF generated using the inverse modeling open-source software described in Li et al. [46]. The Figure shows that the MPC OFF and MPC ON periods have similar levels of daily discomfort. This was examined formally using the Welch's t-test [52] under the null hypothesis that the D_d samples on MPC ON days and MPC OFF days were the same. The test did not reject the null hypothesis at the significance level of $\alpha = 5\,\%$ and therefore supported the observation that the discomfort levels were similar with or without MPC.

4.4. Flexibility metrics

The median values of the hourly load profiles in Fig. 5 are extracted to calculate flexibility KPIs. The two KPIs chosen are Demand Decrease Percentage, DDP, and Demand Decrease Intensity (or Shed Intensity),



(1) Typical cold day (2/14/24) MPC ON

(2) Typical cold day (1/23/24) MPC OFF

Fig. 9. RTU1 example cold day plot. Plot a) shows Outside (green), Supply (orange) and Return (violet) air temperatures. Plot b) shows Supply (orange) air flow rate. Plot c) shows Compressor 1 (orange), Compressor 2 (green), and outside air damper (0 closed, 100 open) control signals. Plot d) shows Zone Temperatures (dashed grey) with Average (red). Plot e) shows UFTs heating valve opening (dashed grey) with Average (red). MPC ON in Figure (1) and MPC OFF in Figure (2). MPC ON is on 2/14/2024 and MPC OFF is on 1/23/2024.

DDI, taken from Liu et al. [53] as a minimum set of KPIs that still provide a comprehensive performance summary. DDP is calculated according to

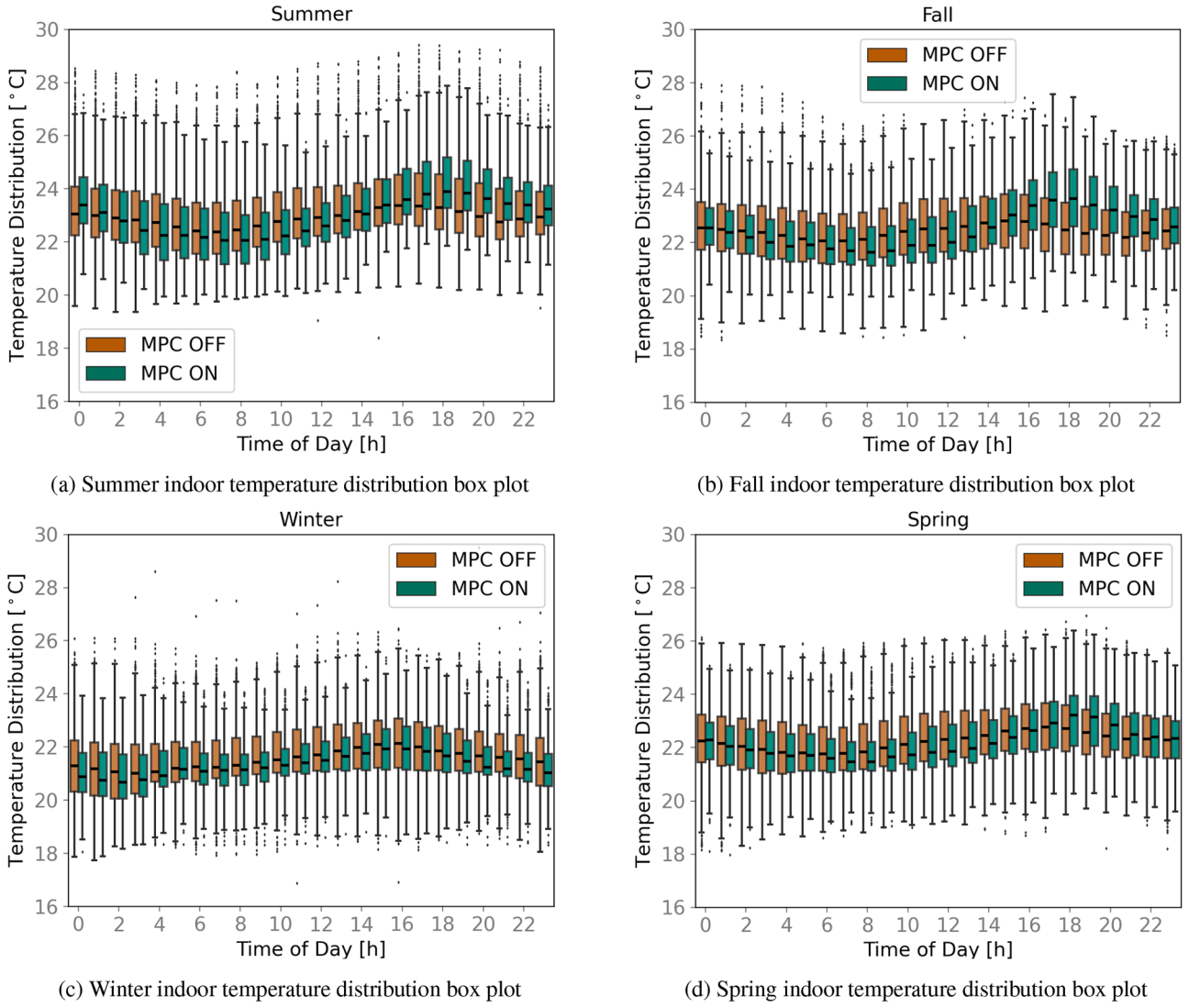
$$DDP = \frac{E_{peak,off} - E_{peak,on}}{E_{peak,off}} \tag{18}$$

where $E_{peak,on}$ and $E_{peak,off}$ are the HVAC energy use for MPC ON and MPC OFF respectively during the n_{peak} hours when the electricity price is highest during the day. In this study, n_{peak} is chosen as 6 h, so that the metric accounts for both electricity price peaks in the Winter season. DDI is calculated according to

$$DDI = \frac{E_{peak,off} - E_{peak,on}}{n_{peak}\, A} \tag{19}$$

where A is the floor area of the building.

Table 3 reports the values of these KPIs. DDP is reported both for the HVAC and whole building, which includes plug loads and lighting, energy consumption, while DDI is reported on for HVAC since it would be almost the same in both cases assuming that the non controllable load is on average the same between MPC ON and MPC OFF periods. Comparing DDP_{HVAC} and $DDP_{Building}$ for all seasons it is clear that the controllable HVAC makes up the majority of the load in the building as highlighted by the small difference between the two KPIs. This is further confirmed by a previous study [30] on B59 time series data for the years 2018 to 2020 that reports HVAC consumption to be between 61 and 88 % of the overall load. Looking at numbers in Table 3 confirms that the MPC is able to shed demand during high price periods in all four different seasons and with different price profiles and the high DDP value compared to the 25 % upper value found in Liu et al. [53] is justified by the high fraction of HVAC controllable load and overall decrease of energy consumption under MPC operation compared to the baseline. In fact it is interesting to notice that DDP_{HVAC} decreases from Winter (71 %), to Spring and Fall (69 % and 64 %), to Summer (49 %), indicating that less demand can be shifted as the seasons get warmer. This is due to the fact that the baseline controller was updated with static pressure and supply air temperature resets to reduce cooling demand, as described in Section 2.2, making the difference between MPC ON and MPC OFF smaller in the cooling dominated days. This is not the case for days requiring heating, as the MPC not only shifts demand, but reduces the consumption for the whole day thanks to a more efficient operation of the HVAC than the baseline, as explained in Section 4.3. DDI is affected in a similar fashion by the difference in heating and cooling demand, as seen by the DDI being 4.4 W/m^2 in the Summer compared to 9.2 W/m^2 in the Winter, which is in range with benchmark



(a) Summer indoor temperature distribution box plot

(b) Fall indoor temperature distribution box plot

(c) Winter indoor temperature distribution box plot

(d) Spring indoor temperature distribution box plot

Fig. 10. Hourly distributions for MPC ON (green) and MPC OFF (red) of indoor temperature for each day of the analysis period: Winter (12/1/23–2/29/24), Spring (3/1/24–5/30/24), Summer (6/1/23–8/27/23), and Fall (9/4/23–11/30/23). Excluded days are as shown in Fig. 4.

range 1–10 W/m^2 presented in Liu et al. [53]. Furthermore, since the average demand of the HVAC is lower in the Summer with respect to the other seasons, the difference between the seasons for the absolute metric DDI is even higher than for the relative metric DDP.

4.5. Commercial building MPC cost savings

Challenge 3 from Section 1.2 suggested that cost savings from MPC providing demand flexibility will depend on the economic incentives to provide flexibility. Therefore, this section introduces a simple method for analyzing cost savings as a function of an economic incentive. To do so, we introduce the electricity price relative standard deviation (EPRSD). The EPRSD is calculated as

$$EPRSD = \frac{\sigma}{\mu} * 100 = \frac{\sqrt{1/n \sum_{i=1}^{n} (p_{ele,i} - \overline{p_{ele}})^2}}{\overline{p_{ele}}} * 100 \quad (20)$$

Where σ is the standard deviation of the electricity price profile, p_{ele}, over the testing period, n, divided by the average price, $\overline{p_{ele}}$, over the testing period. Therefore, an EPRSD value of 0 % means that the price profile is constant, while an EPRSD value of 100 % means that the price can vary such that the difference between the minimum and maximum price is at least twice the average. Fig. 12 shows MPC cost savings for the indicated study's time period as a function of EPRSD for this work from Table 2 along with selected past demonstrations from the literature review in Table 1. EPRSD in this study was calculated under the assumption that the annual TMY3 data for Oakland can be equally divided between Winter, Spring, Summer, and Fall. Then, from the price values for the whole year, EPRSD was estimated for each price scenario, namely $constant$, $scaled$, and $default$. In a similar fashion, EPRSD was calculated for the studies in literature considering the study time period and tariffs used in each period for the respective MPCs. The data points included from past demonstrations are those studies that represented MPC control of a whole-building system and consecutive testing periods longer than five days.

While in Fig. 12 a clear trend can be identified, remaining spread in the data can be attributed to different flexibility potential for the MPC to access and the presence or not of additional economic incentives like demand charges, as discussed in Section 1.2. For example, in the case of Zhang et al. [9], the lower cost savings can be explained with the fact that the savings are based on the whole building energy load, while the HVAC load makes up only between 15 to 25 % of the total. Meanwhile, all other studies report savings in terms of HVAC energy costs. Fig. 12 distinguishes this whole-building energy cost savings with a different shape. In addition, Fig. 12 distinguishes those studies that included de-

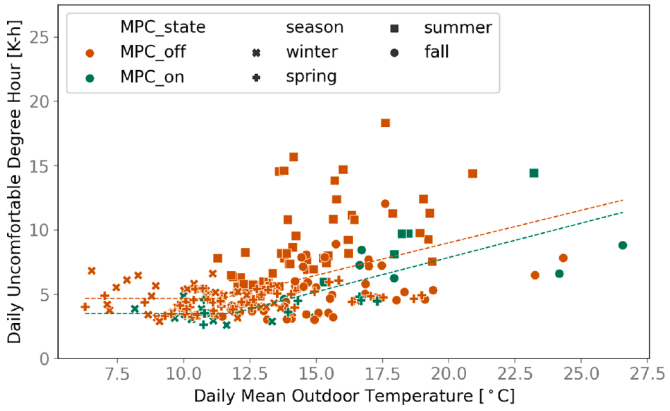


Fig. 11. Daily uncomfortable degree hours plotted against the daily mean outdoor temperature for MPC OFF and MPC ON, with piece-wise linear regression (dashed). Excluded are days as shown in Fig. 4. The days are further divided between the different seasons by using different shapes.

Table 3
Flexibility KPIs calculation based on Fig. 5 data for DDP_{HVAC} and DDI, while $DDP_{Building}$ also includes plug and lighting energy consumption in both MPC ON and MPC OFF periods.

KPIs	Winter	Spring	Summer	Fall
DDP_{HVAC} [%]	71	69	49	64
$DDP_{Building}$ [%]	67	65	41	56
DDI [W/m^2]	9.2	7.1	4.4	8.7

mand charge reductions in the MPC with a different color. Lastly, a difference in flexibility potential can be seen by the point that corresponds to Kim and Braun [27]. With an EPRSD of about 60 %, the cost savings is about 30 %, lower than the 50 % of the scaled scenario of this paper's study. This difference is supported by the fact that the authors in Kim and Braun [27] pointed out that the building under analysis has poor insulation and lack of thermal mass to activate. Instead, in the present study, the building is fairly well insulated and the UFAD system allows activation of the concrete core of the building to have a larger effective thermal mass to shift the load.

Despite the spread in results, the trend shown in Fig. 12 implies that providing a higher flexibility incentive gives the MPC more potential savings to achieve with respect to a baseline. However, the sample size of building types and climates in literature is still small and only few studies allow for estimating cost savings over reasonable and representative time periods. For this reason, additional tests with different pricing conditions, climates, building types, and HVAC types should be tested and reported to help create a regression one can use in the future to rapidly estimate savings potential and determine if MPC can be a cost-effective retrofit investment for a given situation. Furthermore, Fig. 12 could be created considering different flexibility metrics on the y-axis, such as the ones calculated in Section 4.4, to analyze the flexibility performance as function of the economic incentive. However, currently there is no consistency in the MPC literature on the reporting of these flexibility KPIs.

Fig. 12, especially with more data from additional case studies, allows a building owner or operator to analyze their own building's dynamic electricity price profile to determine its EPRSD, use the trend line to get a preliminary, data-informed estimate of potential percentage cost savings achievable with MPC, and combine this estimated savings potential with their own quotes for MPC implementation (commercial software, installation, commissioning) and their specific energy expenditure to perform a more relevant and tailored payback calculation for their unique situation.

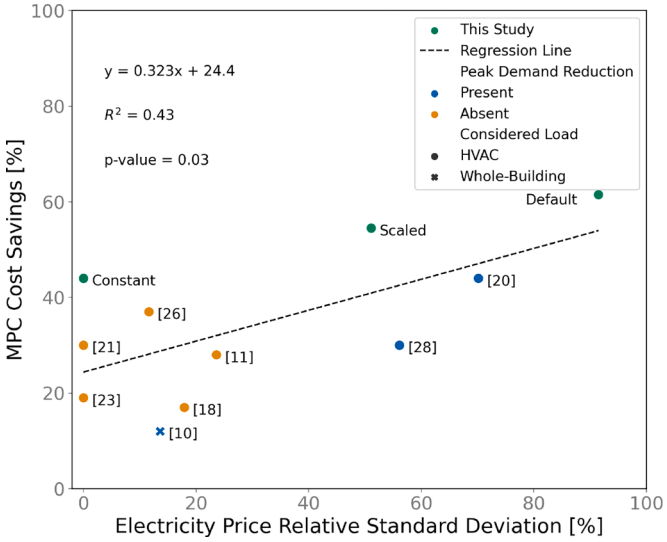


Fig. 12. MPC cost savings of studies presented in Table 1 and results from Table 2 plotted against electricity price relative standard deviation (EPRSD). Appendix B reports how scaled and constant scenaros were obtained from the default scenario. Colors indicate if the study included demand charges in the MPC. Shape indicates if the savings is with respect to only HVAC energy or whole-building energy. Labels indicate reference study.

5. Discussion

5.1. Field test results

The field test was extended from Zanetti et al. [29] by conducting two additional tests, resulting in a total of four two-week tests across all four seasons. This provided sufficient data, comprising 30 days for the Model Predictive Control (MPC) and 187 days for the baseline, spread throughout the year to train a change point regression model based on daily energy consumption and external temperature for both the MPC and baseline controllers. The resulting models were used to estimate the potential typical year MPC savings by applying the model to Oakland TMY3 weather data in three different pricing scenarios: the default highly dynamic price, a scaled price with a lower peak-to-trough ratio, and a constant seasonal price. Although using the MPC solution found with the default highly dynamic price to estimate operational costs with different price profiles is an approximation, as the MPC would have yielded a different solution, it is still useful to gain insight into the savings potential in each scenario. The final cost savings ranged from 44 % in the constant price scenario to 61 % in the highly dynamic price scenario. These results indicate that the MPC was able to leverage the building's thermal mass to shift the load, as demonstrated also by the flexibility KPIs, where the Demand Decrease Percentage (DDP) ranged from 49 % in summer to 71 % in winter. The MPC also significantly reduced energy consumption, particularly in winter, by better managing the RTUs' air supply. Specifically, the MPC consistently attempted to reduce flow rates to minimize fan power consumption while maintaining minimum outdoor air change and minimum supply air temperature requirements, especially in winter. In summer, it performed extensive night precooling when the outside air was cooler than the indoor environment. We argue that without full supervisory integration into the Building Automation System (BAS), the MPC would not have been able to optimize RTU operation to this degree, and tweaking only thermostat set points would have resulted in significantly lower savings. Finally, the energy consumption reduction and shift did not come at the expense of occupant comfort, as shown by the p-value analysis rejecting the null hypothesis that the MPC increased discomfort.

5.2. Implementation effort

While the MPC implementation in this study was effective in shifting energy and reducing cost, updating the software from Blum et al. [16] still required significant effort, although not specifically on updating the model or optimization problem, which were relatively straight-forward. Rather, the effort was spent more on data and BMS integration tasks. The effort was quantified as approximately 88 person-days or four person-months of work, corresponding to 35 % of the effort spent in Blum et al. [16] to develop the MPC from scratch, which did not include the controller software MPCPy [35] development. This effort and subsequent division in tasks were estimated using the employee time sheets, commit history of the software repository, and meeting notes. Approximately 50 % of the time was spent by a new developer in learning the code base and integrating new and replacing data streams as described in Section 3.1. As already mentioned in Blum et al. [16], increasing the number of underlying data sources and streams raises the chances of failures and data interruptions. The remaining 50 % of effort was spent in a variety of tasks. A first was monitoring the data quality for the streams necessary to properly operate the MPC, especially during testing periods. A second was checking the validity of the MPC results before field tests. A third was updating and testing the MPC-ALC interface as described in Section 3.1. A fourth was addressing unforeseen circumstances, such as modifying the MPC to incorporate a new mode of operation for wildfires, which could be activated when facility operators are required to eliminate outside airflow. A fifth was the fact that BMS control logic and configuration had to be updated to allow MPC to turn RTUs on during unoccupied hours, without repercussions on other parts of the control logic. This was only possible through close collaboration with facility managers.

With this effort required to update the MPC, it is worth mentioning that it took facility managers an estimated 19 person-days to implement and test the reset logic based on occupancy schedule and zone heating and cooling requests explained in Section 3 to reduce energy use after the findings of the first MPC implementation [16]. When thinking about enabling demand flexibility, it would likely require a larger effort to mimic and implement the different ways MPC achieved flexibility with rule-based logic in the BMS, in addition to capturing energy savings potential during heating. Furthermore, these rules would only be valid for a specific price and season they are written for.

A final consideration regarding the effort involved in implementing MPC is the substantial time and resources required to access, understand, and set up the underlying data and control infrastructure in buildings before MPC can be deployed. The use of Energy Management and Information Systems (EMIS) can mitigate these challenges and create a pathway for deploying MPC. For instance, recent work showcased how EMIS platforms with advanced two-way communication can enable automatic fault correction, supervisory controls, and demand flexibility [54–57]. These capabilities could support continuous supervisory control, like MPC, in systems with clearly defined data streams and fully commissioned HVAC systems. Additionally, EMIS can facilitate fault detection and diagnostics in Building Management Systems, improving system operations and achieving average portfolio savings of 9 % with a two-year payback period [58]. On the other hand, MPC developers need to understand the underlying control logic. To facilitate this task, MPC implementers could leverage the OpenBuildingControl project and related ASHRAE Standard 231 [59]. These initiatives aim to provide a standardized and digitalized control description language, making it easier to process and understand building control sequences.

5.3. Limitations of the study

While this study provides valuable insights, it is important to acknowledge its limitations. Firstly, the findings are derived from a comprehensive study of a single large commercial building with a specific Underfloor Air Distribution (UFAD) system in a mild climate. Therefore, the direct transferability of the exact savings percentages and flexibility KPI values to other building typologies, HVAC configurations, or climatic regions should be approached with caution.

Secondly, the implementation effort quantified (four person-months) pertains to the extension and operation of an existing, research-grade MPC system. This effort does not represent the full lifecycle cost of developing an MPC from scratch or deploying a commercial, off-the-shelf product, which would involve different cost structures and potentially different expertise requirements. The unique nature of large commercial buildings also means that the scalability of this specific solution, without further adaptation or leveraging more standardized platforms (as discussed in Section 5.2), is not inherently guaranteed.

Thirdly, our economic analysis, while providing strong indicators of potential savings against price variability, does not offer a definitive payback period for this specific research deployment. Such a calculation would depend on commercial product costs, specific installation expenses, and prevailing economic incentives not captured within this research scope. Instead, we provided a framework (EPRSD analysis) to aid stakeholders in making such tailored assessments. Furthermore, when estimating annual cost savings under the alternative (scaled and constant) price profiles, the MPC control strategy optimized for the default dynamic prices was utilized, representing an approximation in our multi-scenario analysis. A fully re-optimized MPC for each distinct price profile might yield slightly different savings.

Finally, the meta-analysis of cost savings versus EPRSD (Fig. 12), while indicative of a positive trend, is based on a limited number of comparable published studies. A larger and more diverse dataset from future demonstrations would be beneficial to strengthen the robustness and generalizability of such correlational analyses.

6. Conclusions

Previous studies have demonstrated the potential for MPC to provide demand flexibility in different types of buildings or testbeds, and have recently started quantifying implementation effort, cited as a key barrier to widespread adoption. The value of providing demand flexibility could justify implementation costs, however, the value provided for a particular building depends on economic incentives to provide flexibility and the accessibility of the MPC to load flexibility. Furthermore, there remains a lack of studies demonstrating such value in real commercial buildings. Therefore, this study demonstrated MPC controlling a large commercial building HVAC system to provide demand flexibility in response to four different electricity price profiles in four different seasons, estimated annual cost savings and demand flexibility metrics, estimated implementation effort for extending a previous MPC that focused on energy minimization to the one used in this study that enabled price-responsiveness through cost minimization, and compared cost savings to those reported in literature in terms of economic incentive, quantified by the electricity price relative standard deviation (EPRSD).

The first conclusion is that MPC successfully provided load shifting in response to four different dynamic electricity prices in four seasons, without compromising thermal comfort. This resulted in an estimated annual cost savings of approximately 61 % and energy savings of approximately 45 %. The energy savings were driven primarily by more efficient heating provided by the MPC than the baseline control, while added cost savings were driven by shifting load away from periods of high prices during both cooling and heating. The estimated demand decrease intensity, DDI, ranged from 4.4 W/m^2 in the Summer to 9.2 W/m^2 in the Winter.

A second conclusion is that while the effort required specifically to change the MPC from energy minimization to price-responsive cost minimization was low, including just needing to update price data from test to test, other issues continued to require significant effort. Overall, extending the MPC from a previous effort [16] to this one and executing

the field demonstration required an estimated four person-months, or 35 % of the previous implementation effort. This included effort for a new developer to learn the code base, update deprecated data streams, implement new logic in the MPC and ALC to enable MPC control during unoccupied periods, and monitor MPC data streams during tests. This stresses the importance of implementing robust data communication for MPC, which is used for parameter estimation, state estimation, and control. To reduce maintenance effort at scale, robust data pipelines and platforms with alarm systems and watchdogs are necessary, and if possible, redundant data streams to increase reliability. In addition, it is important to keep in mind that the effort required to implement HVAC resets based on occupancy and cooling requests in the conventional logic was also significant, and does not include the energy savings the MPC achieved during heating, nor the demand flexibility in response to four different prices signals within each season. Implementing such flexible control for these objectives in the BMS logic would also require significant effort, and still may only be valid for the combination of pricing and seasonal conditions observed.

A third and final conclusion is that the value of MPC providing demand flexibility, quantified by the energy cost savings, depends on the economic incentive. Plotting cost savings resulting from this study, including a sensitivity to price profile, and from previous studies as a function of EPRSD shows a positive correlation from 25 % savings at 0 % EPRSD to 54 % savings at 100 % EPRSD. However, consideration of actual cost savings must also be given to the effects of MPC's accessibility to energy storage, the fraction of controlled load to total load, and the presence or not of additional economic incentives like demand charges.

We believe that future work on MPC can be divided into two key areas: implementation cost reduction and scalability, and additional value propositions.

Regarding implementation cost reduction and scalability, several avenues of research are promising. Demonstrations of MPC in various building types, climates, and economic contexts would provide valuable data on operational cost savings, enabling more accurate estimates of expected savings so as to be able to generate implementation cost targets for acceptable payback periods. The development of fault-tolerant MPC guidelines and frameworks is also crucial, as these would allow systems to continue operating effectively even in the event of component failures or data losses. Furthermore, the use of semantic data models could facilitate the integration of different building systems, reducing the complexity and cost of implementing MPC. Creating templates that are representative of MPC problem formulations for different HVAC systems can also help reduce implementation time and complexity. Lastly, we acknowledge that one of the main challenges to implement MPC in existing buildings is gaining access to, understanding, and commissioning the underlying data and control infrastructure, which can be extremely time consuming. Therefore, any research going into reducing the access barrier to high quality data and ease of access to control points in BMS will greatly help making MPC more scalable and cost effective.

In terms of additional value propositions, several research opportunities remain to be explored. One key area is promoting greater interaction between buildings and the grid, where MPC could optimize building load shapes in response to grid signals and provide the grid with accurate day-ahead predictions of building loads. Another area is resilience, where MPC can help optimize building operation and mitigate the impact of extreme events, both at the building level, to maintain minimum levels of comfort and operation, and at the grid level, to withstand and recover from disruptions. Finally, incorporating indoor air quality and humidity considerations into MPC formulations can help reduce or dynamically manage ventilation and dehumidification energy use, a significant portion of the HVAC load, while maintaining or even improving indoor air quality and occupant health.

CRediT authorship contribution statement

Ettore Zanetti: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **David Blum:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Hongxiang Fu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Formal analysis, Data curation; **Chris Weyandt:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation; **Marco Pritoni:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization; **Mary Ann Piette:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Funding acquisition, Conceptualization.

Data availability

Data can be made available up request depending on type of data and in agreement with LBNL data sharing policies

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used cborg (https://cborg.lbl.gov/) deep thought in order to help improve English in some paragraphs. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Additional data

The bottom plot in Fig. A.13 shows that the assumption made in Eq. (12) is reasonable. When the HVAC panel consumption is close to the cutoff threshold, between 7 and 10 p.m., $P_{hvac1} + P_{hvac2}$ is very close to p_{fantot}, indicating p_{fantot} is representative of the overall HVAC consumption when the measurement at the breaker is not available. This holds true for several days of the year where this similarity was double checked by the authors.

Fig. A.14 shows the cost scatter plots for the scaled and constant price cases. The piece wise linear models were then used to find the results shown in Table 2.

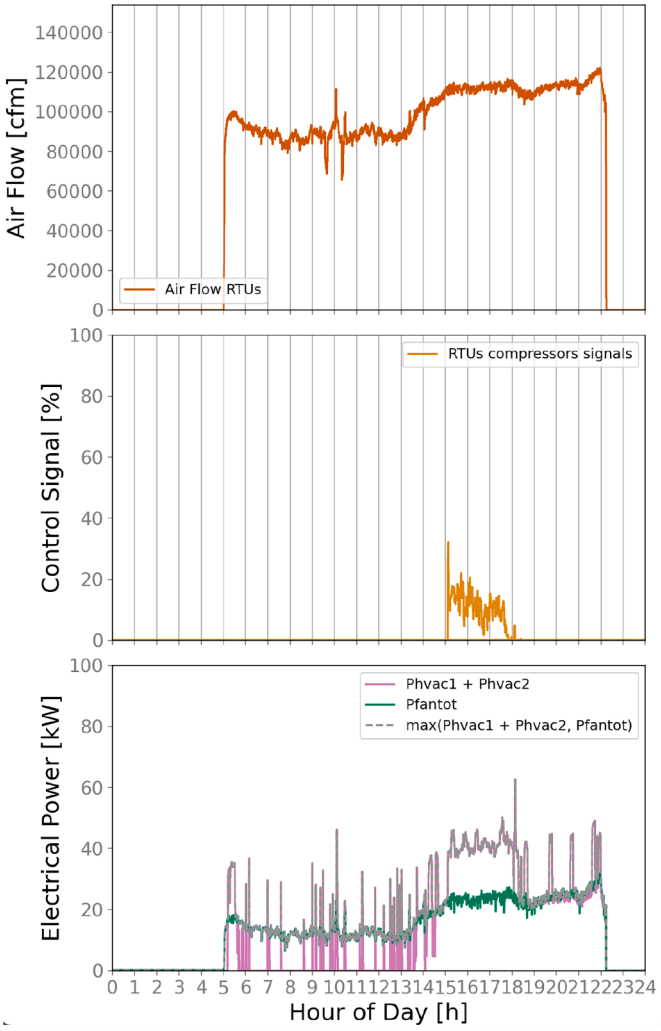
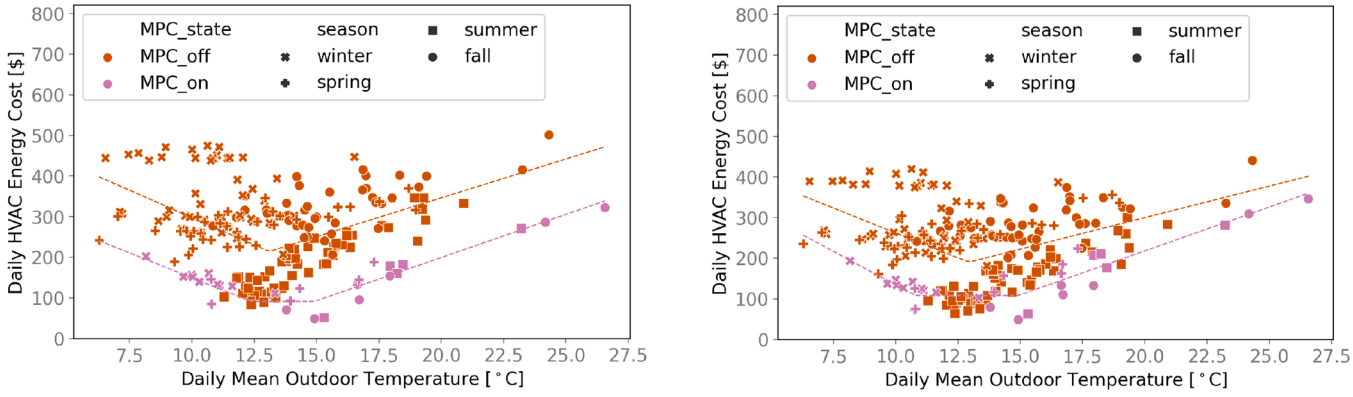


Fig. A.13. Daily hourly plot reporting sum of all RTUs supply and return air flows (top), all RTUs compressors signals (middle), and HVAC electrical powers (bottom).



(a) HVAC Energy Cost scatter plot with scaled prices

(b) HVAC Energy Cost scatter plot with constant prices

Fig. A.14. Figure a) and b) show cost plotted against the daily mean outdoor temperature for the scaled and constant prices shown in 7. Excluded are days as shown in Fig. 4. The days are further divided between different seasons by using different shapes. Piece-wise linear regression lines for MPC ON and MPC OFF (dashed lines).

Appendix B. Seasonal price calculation

The scaled prices shown in Fig. 7 were calculated following a minimization optimization problem with the objective of reducing the peaks to trough ratios:

$$\min_{p(t)} \quad J = \left(\int_0^{24h} ((p(t)')^2 + (p(t)'')^2) dt) + \int_0^{24h} p(t)_{def} dt - \int_0^{24h} p(t) dt \right)^2 \tag{B.1}$$

$$p(t) \leq \frac{max(p(t)_{def}) + p_{cost}}{2} \tag{B.2}$$

Where $p(t)$, $p'(t)$,and $p''(t)$ are respectively the electricity price, its first, and second derivatives for each hour of the day initialized using the default price $p(t)_{def}$. The objective of the minimization problem is optimizing $p(t)$ such that the area under curve remains as close as possible to the default $\int_0^{24h} p(t)_{def} dt$, while preserving the shape of the curve, and $p(t)$ has to be lower or equal to the average between the maximum value of the default price $max(p(t)_{def})$ and the constant price p_{cost} calculated as $\int_0^{24h} p(t)_{def} dt$ divided by the 24 h of the day.

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